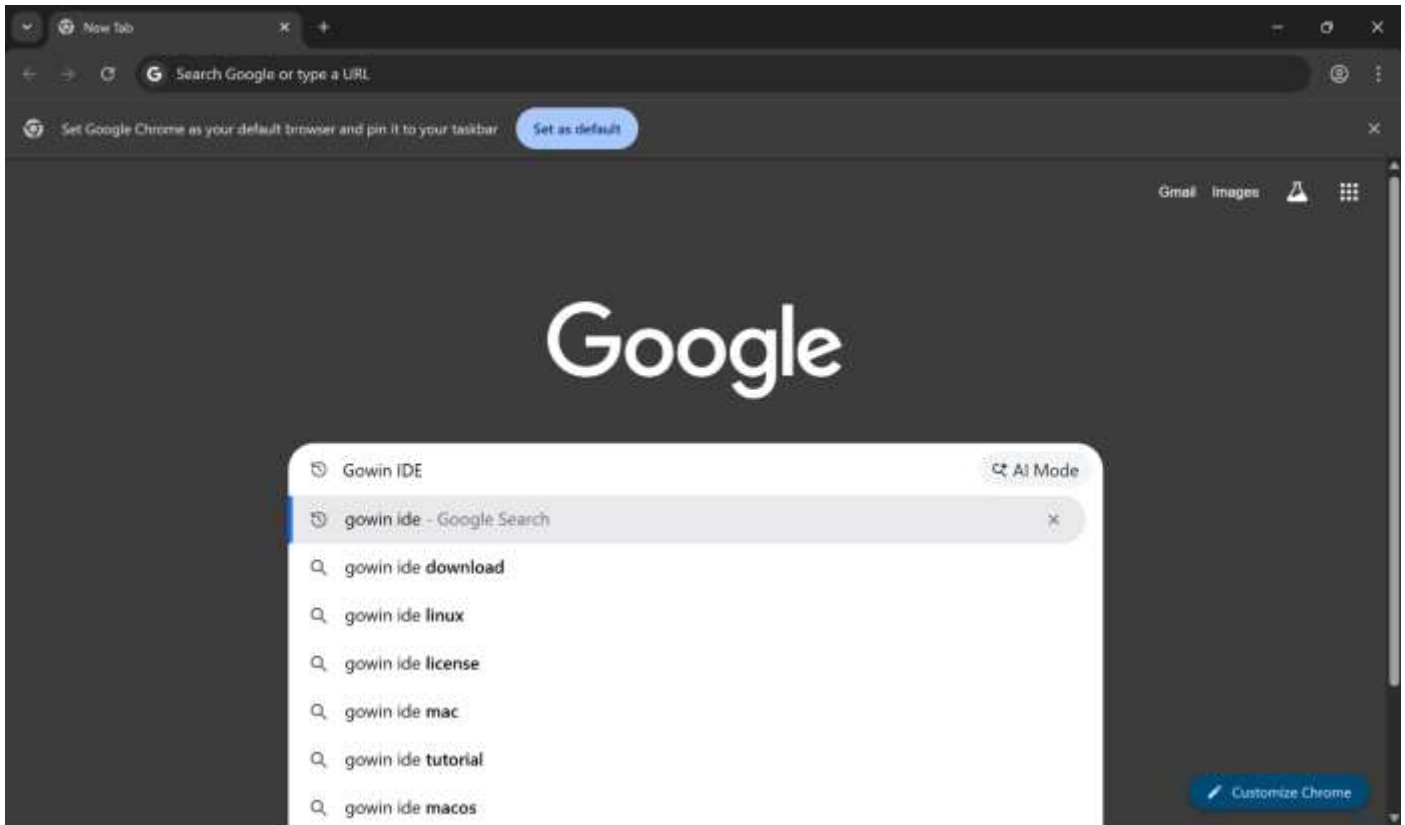
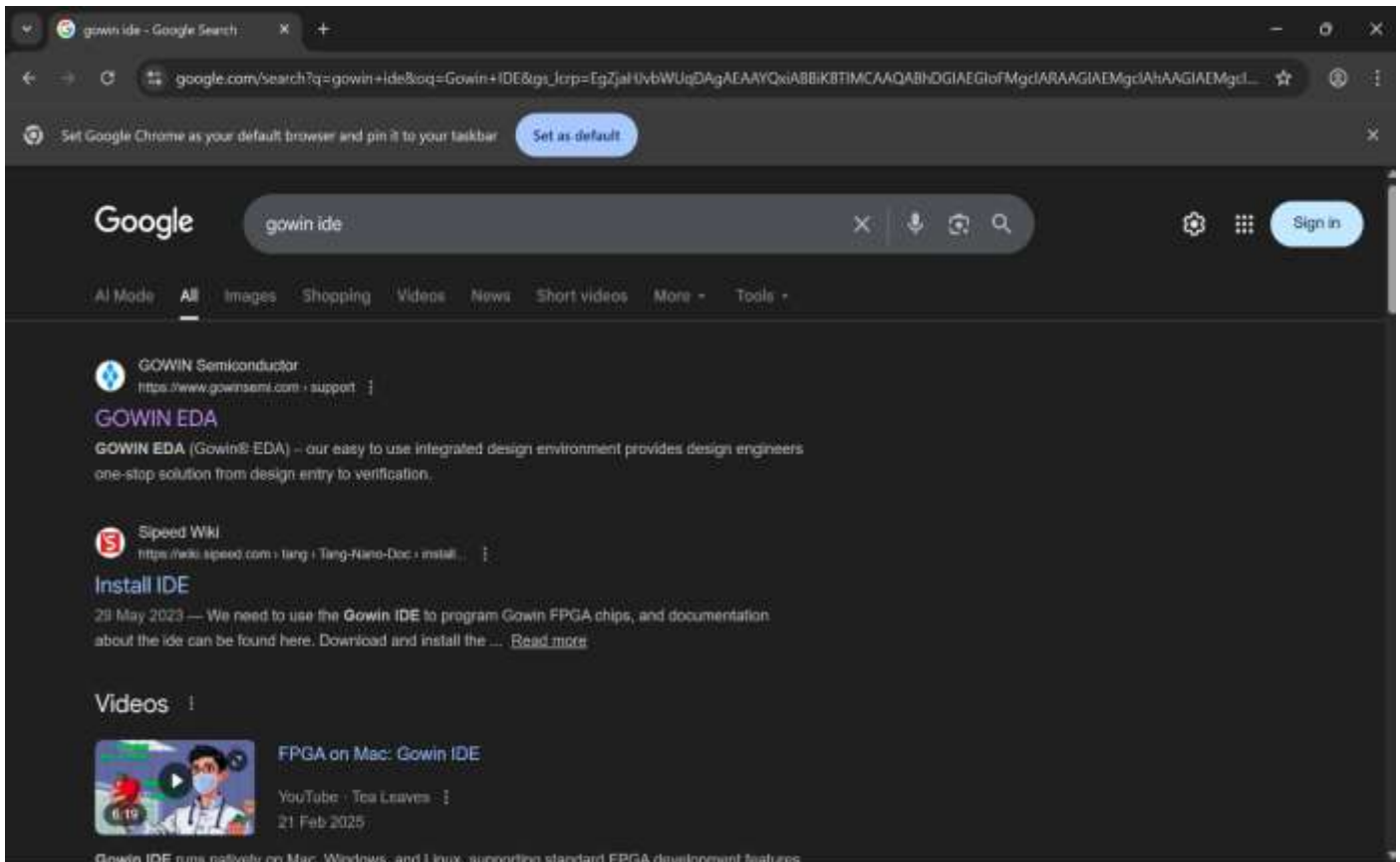


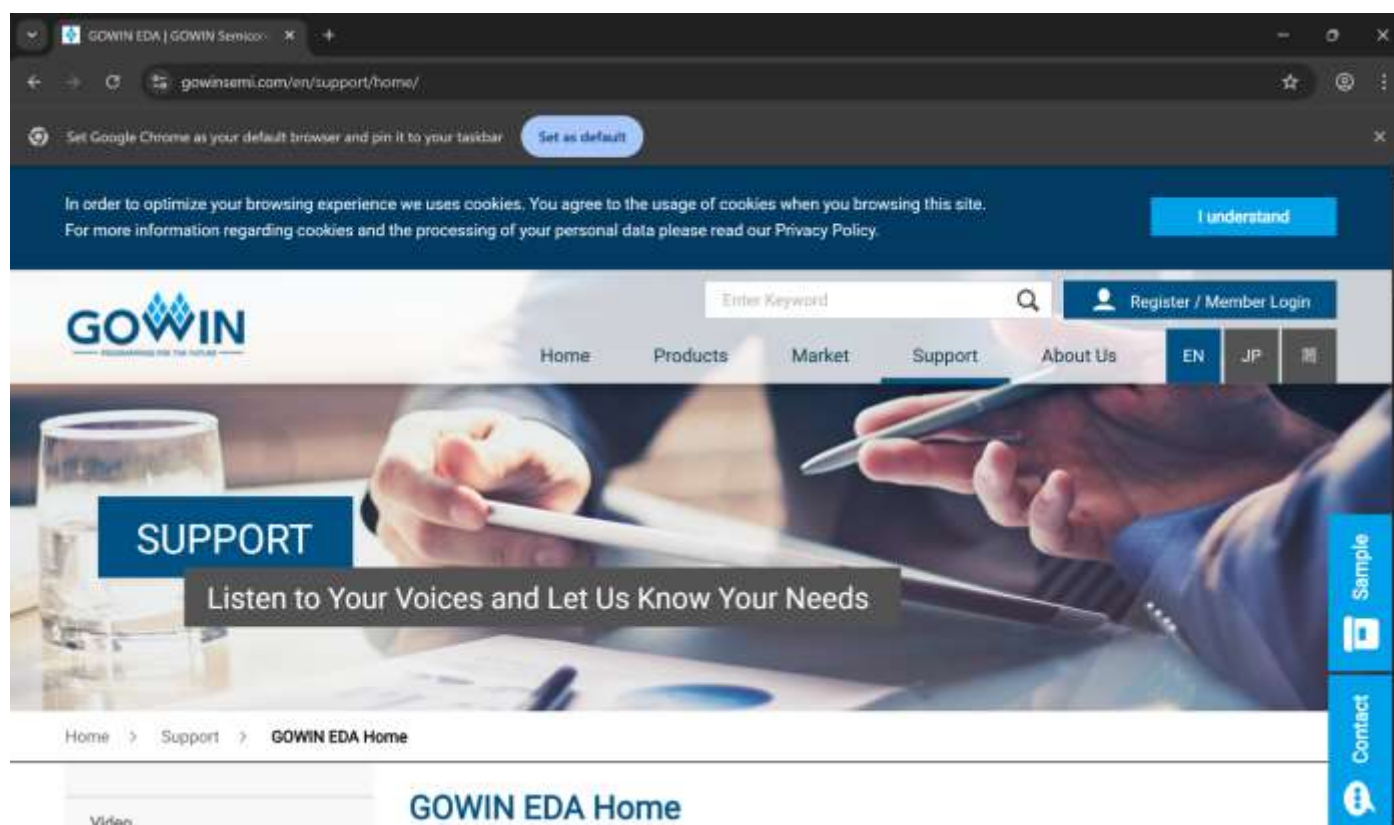
CHAPTER 4 – SOFTWARE DESCRIPTION



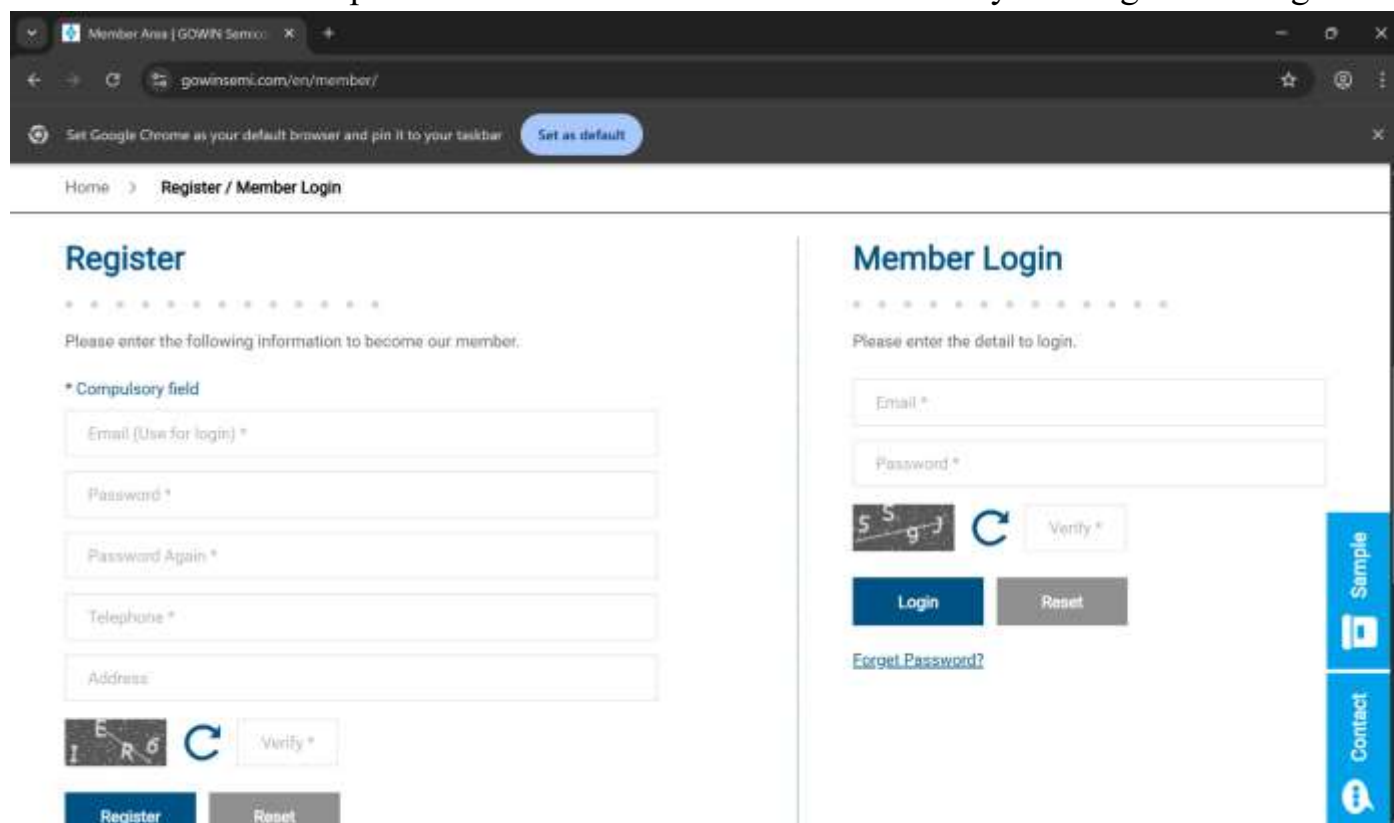
Open the google and type GOWIN IDE in search bar.



After searching GOWIN IDE ,click on the first site that is on the top.

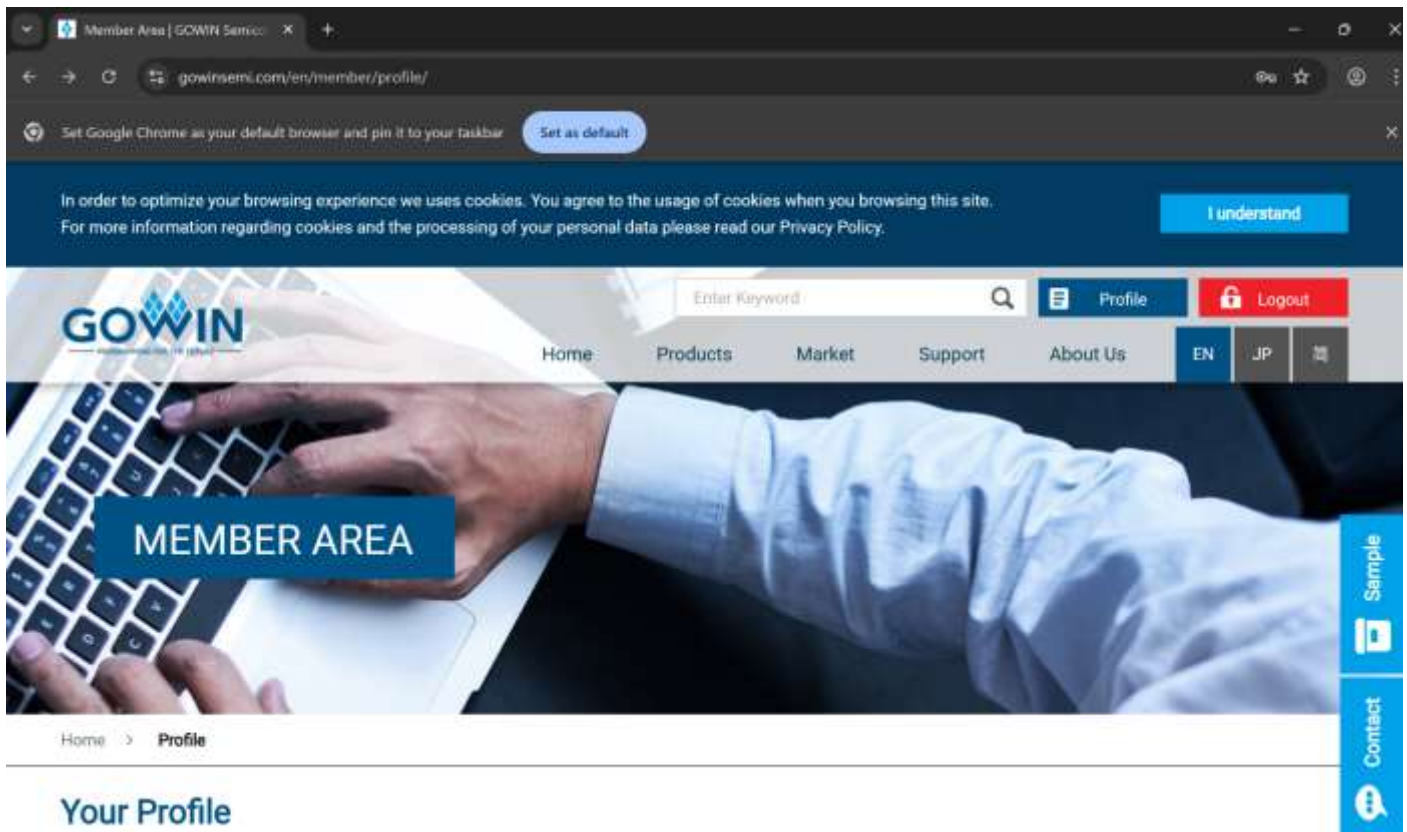


GOWIN IDE will be opened like this we have to create account by clicking on the Register.

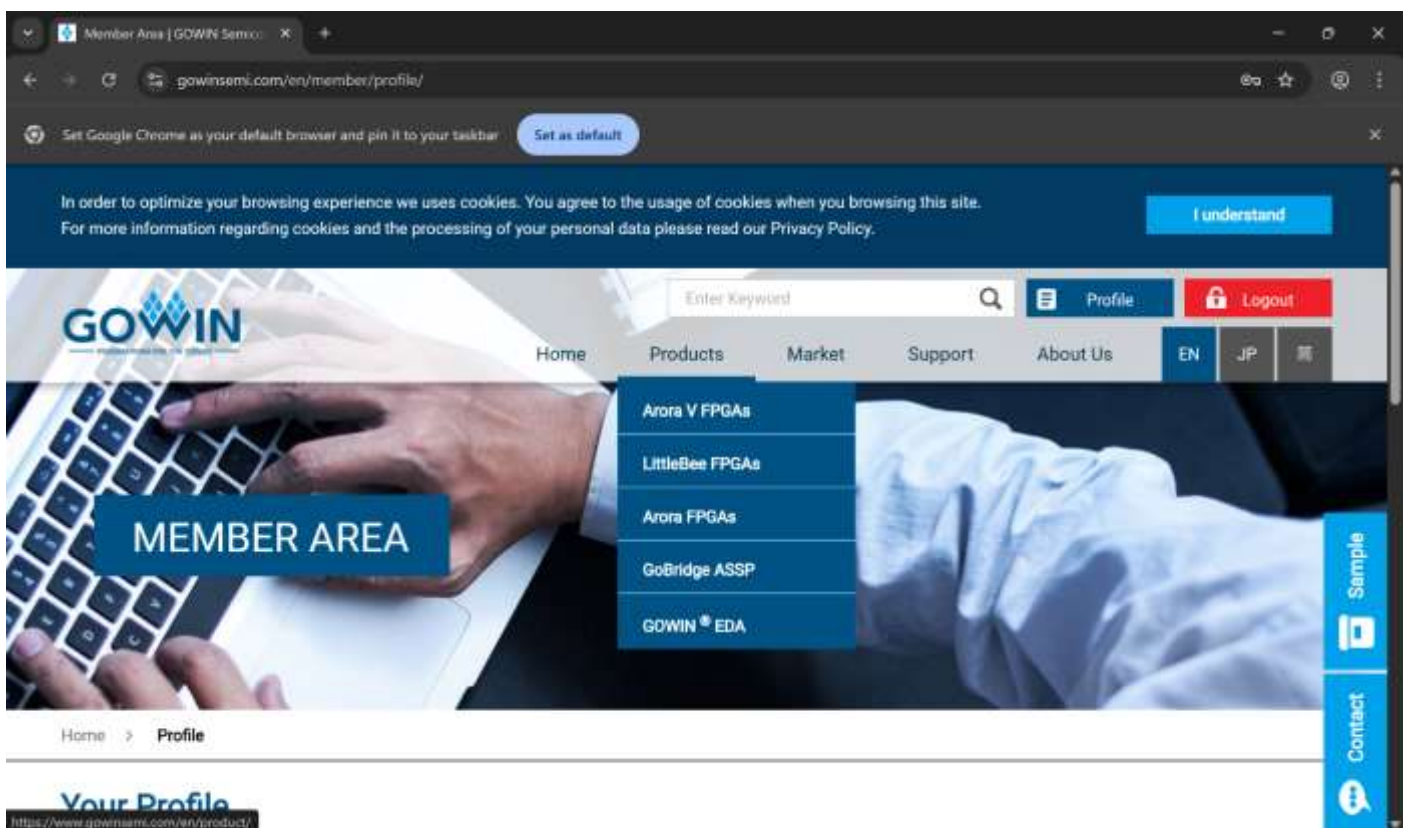


Or if there is any account use member login.

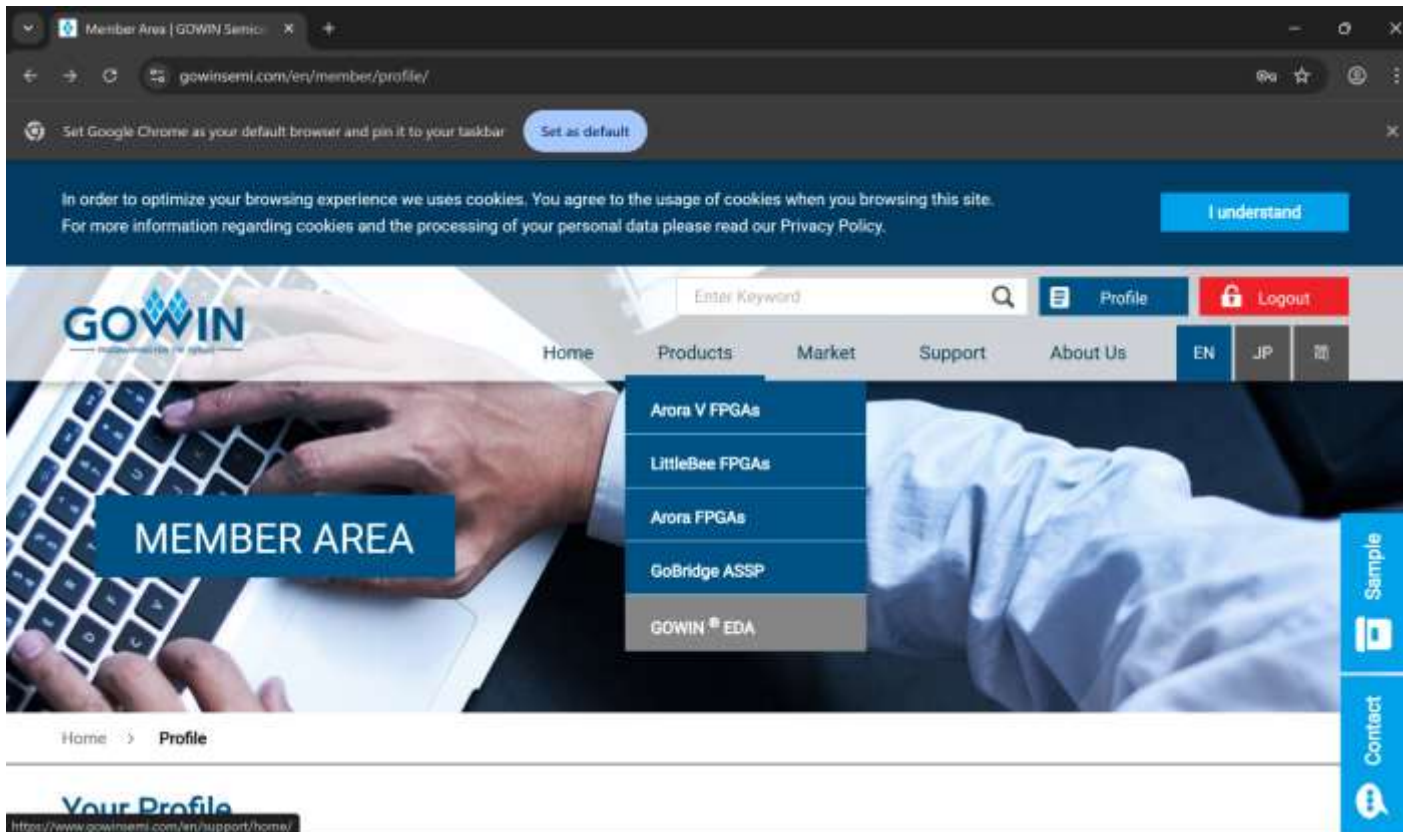
Here, we need to create account or we have to login if we already have an account.



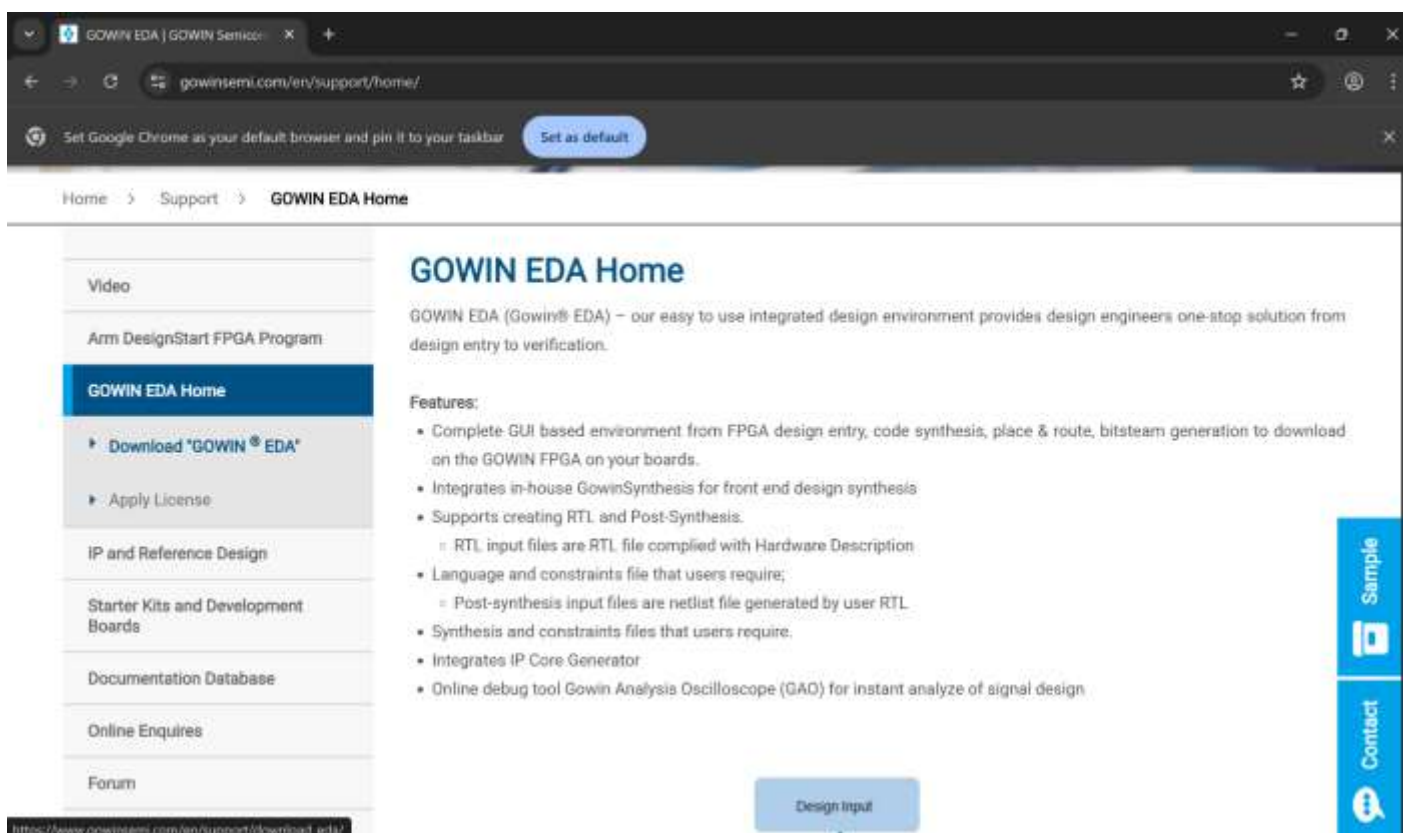
GOWIN IDE will be opened as shown in the above photo.



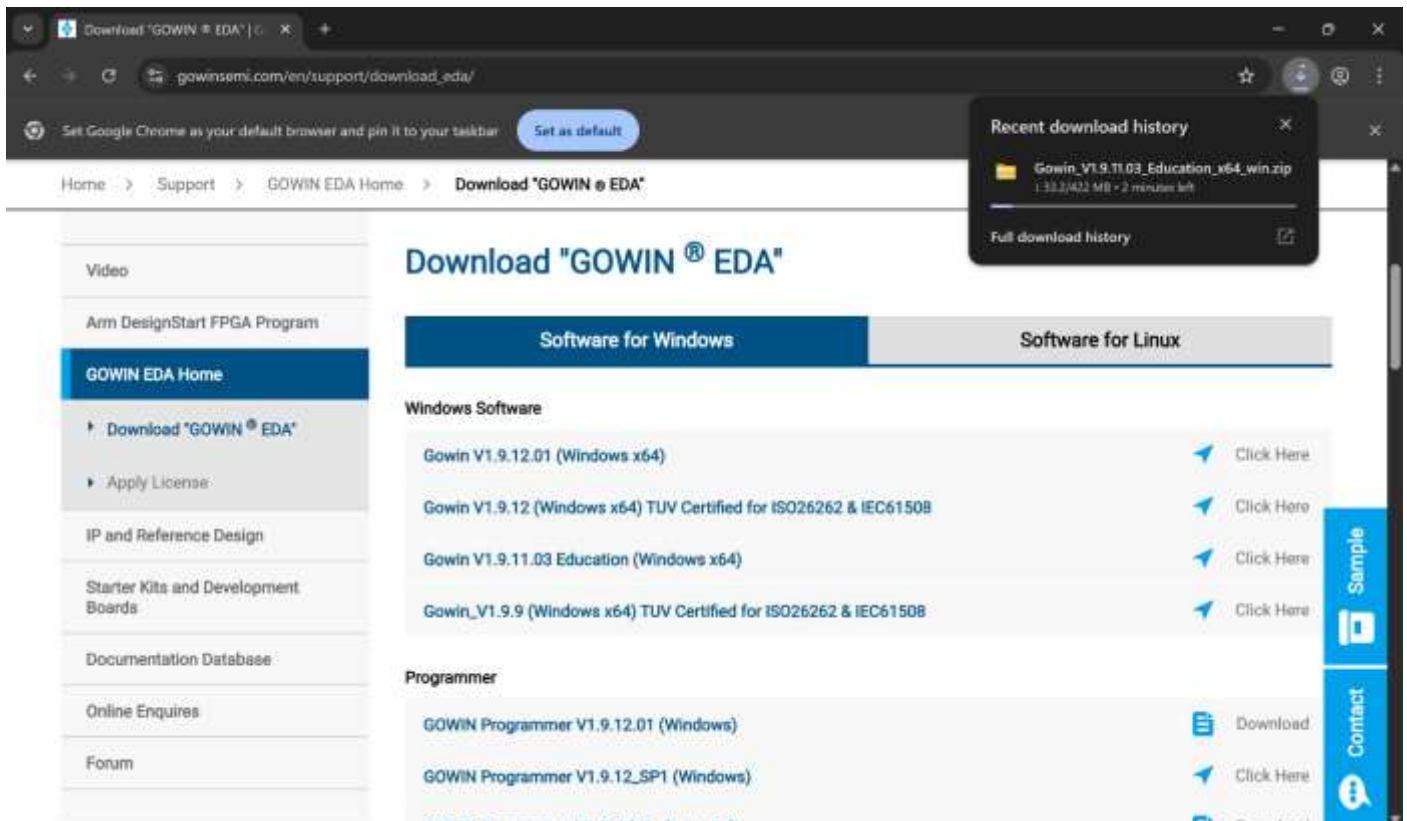
Click on the PRODUCTS option in the, after clicking the products we will see some options as shown in the above photo. Now click on the GOWIN EDA.



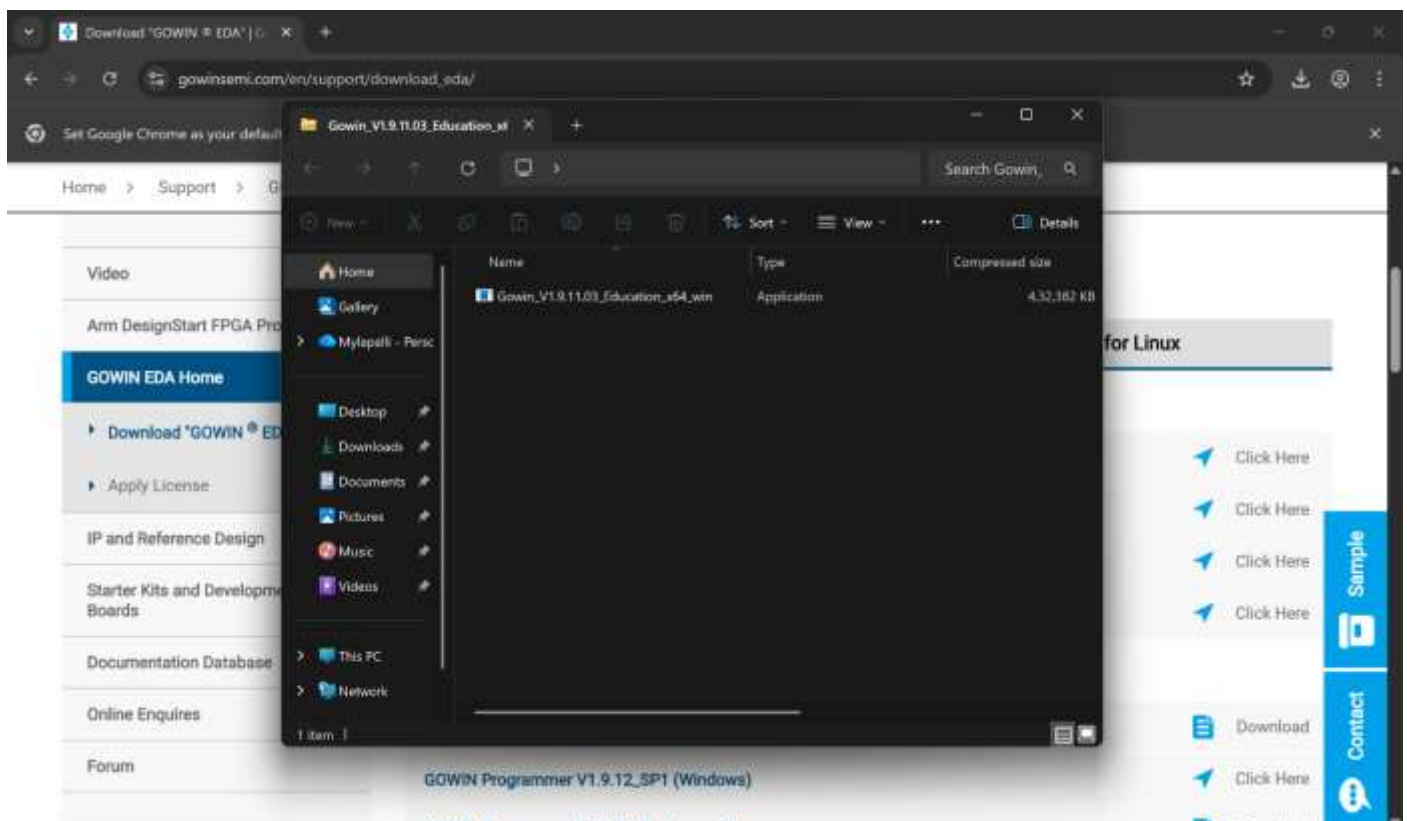
Click on the GOWIN EDA option.



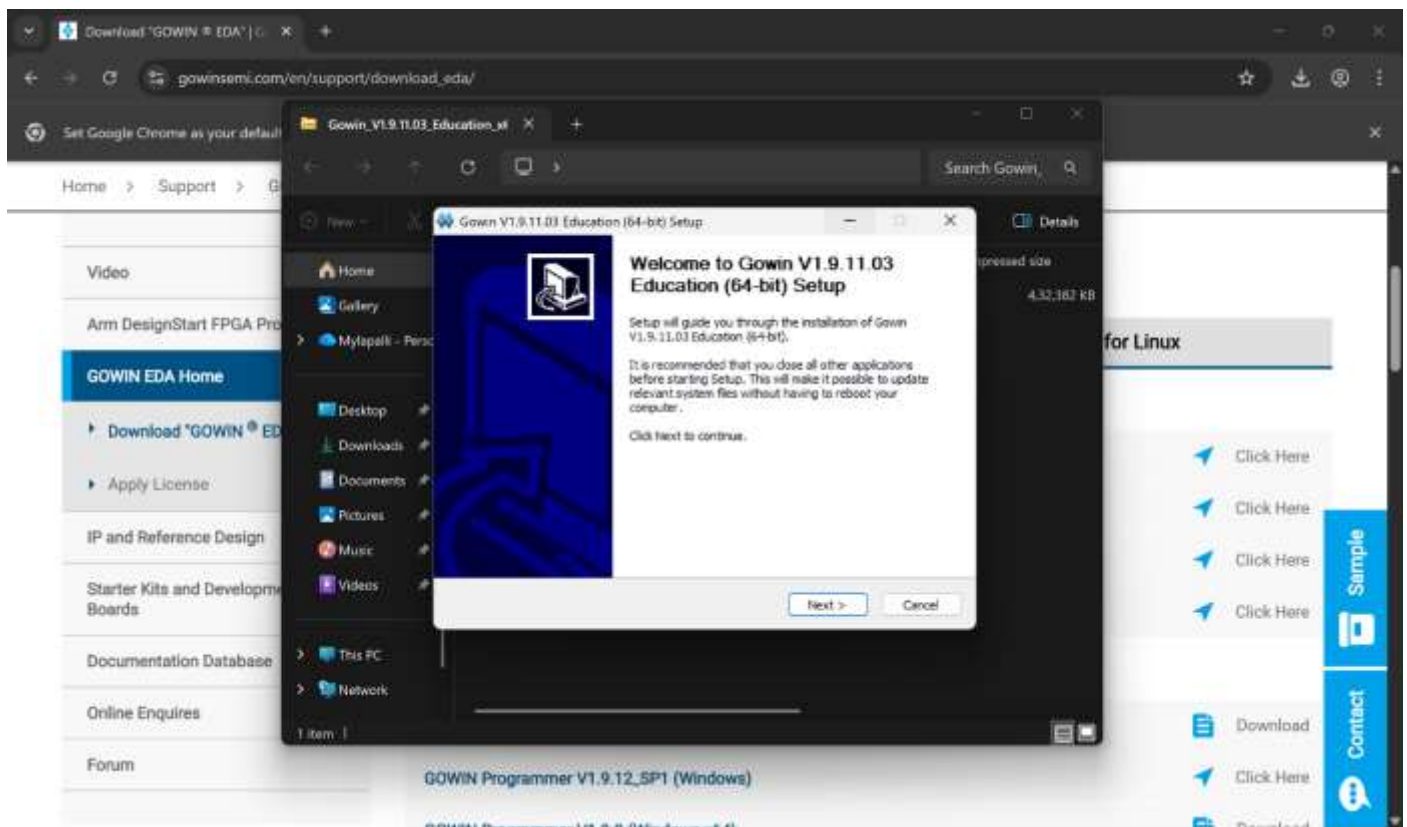
After clicking the GOWIN EDA, we need to click on the Download GOWIN EDA option in it .



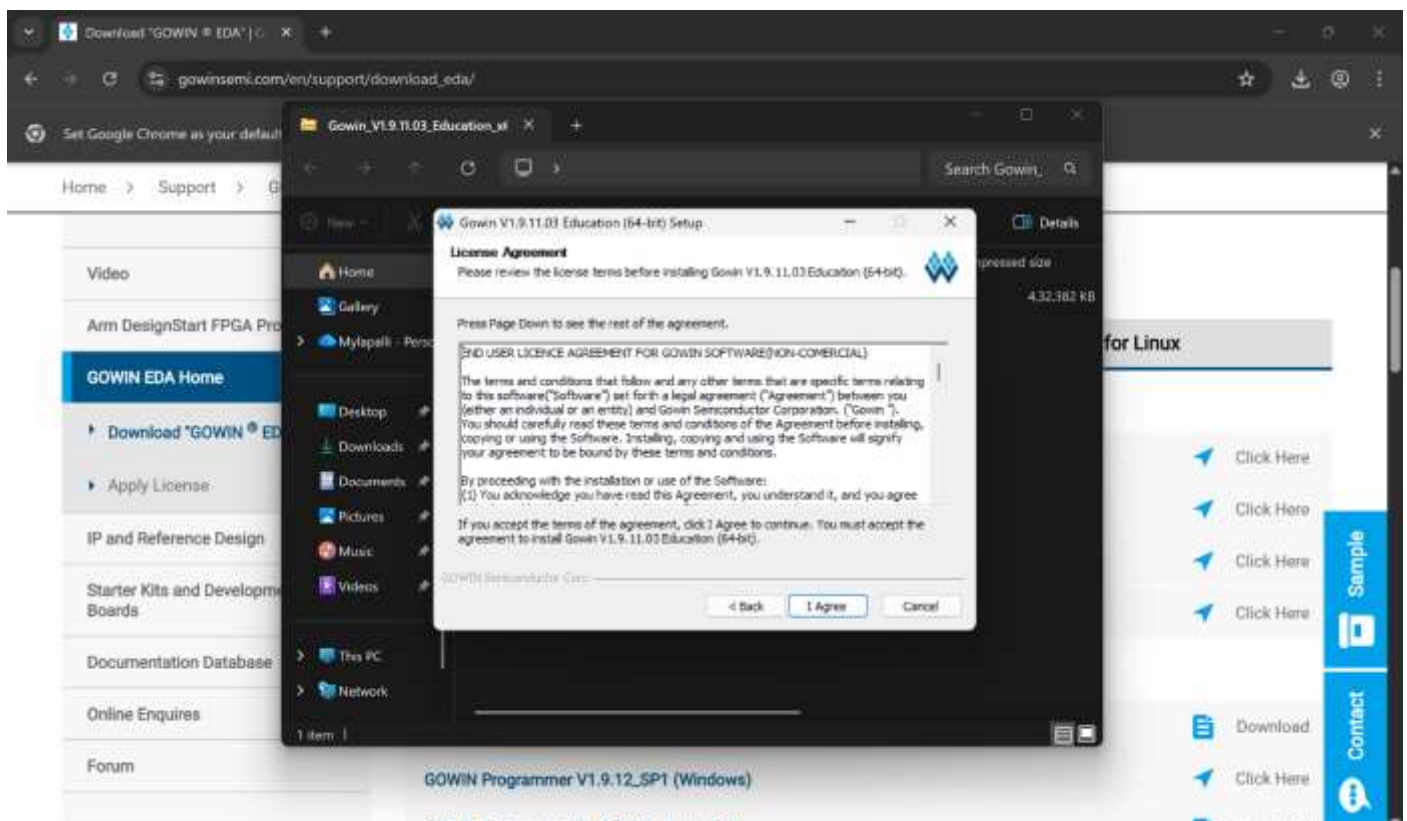
After clicking the Download GOWIN EDA, we will see different windows software and different programmers, click **Gowin V1.9.11.03 Education(windows x64)**



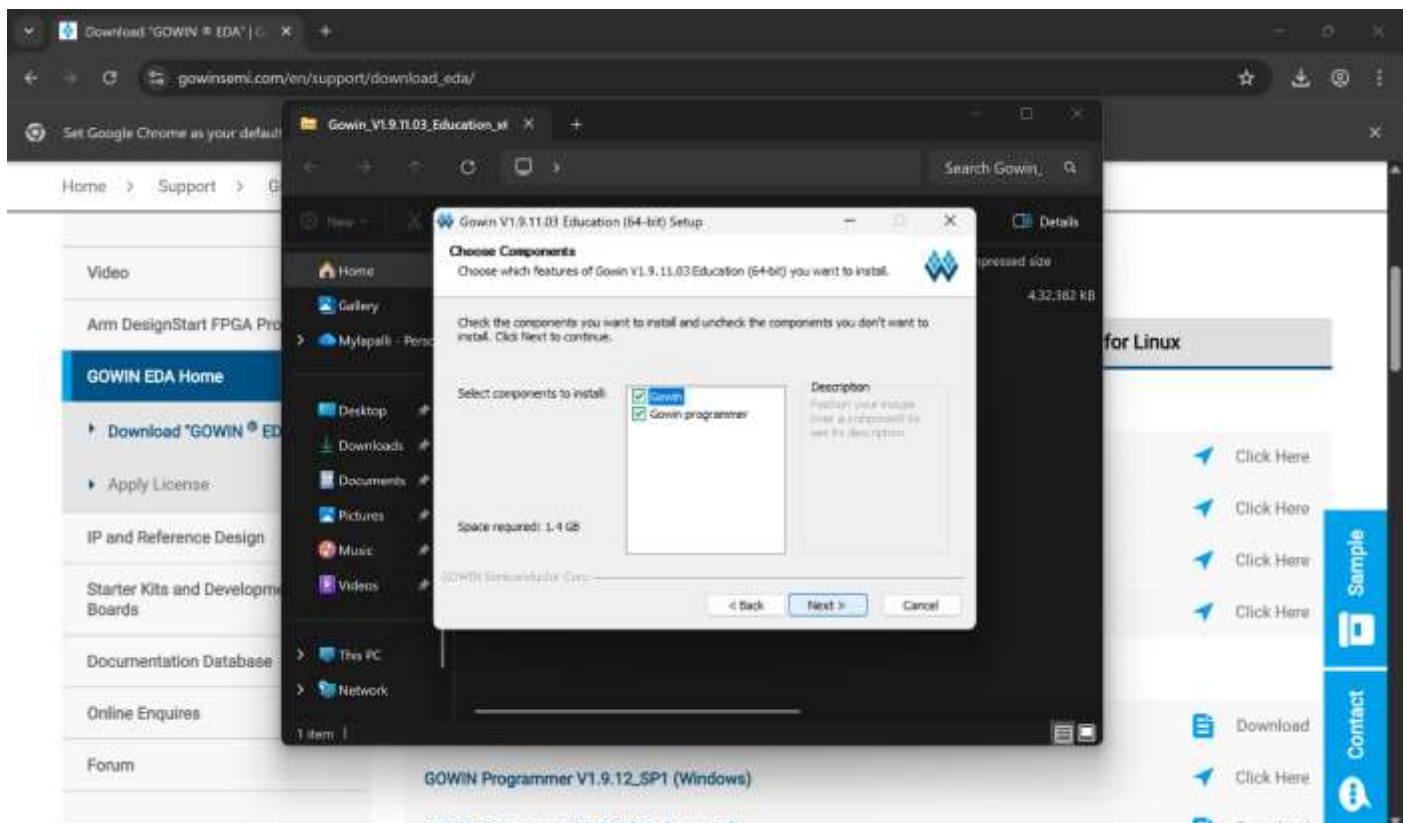
After installation, open the application



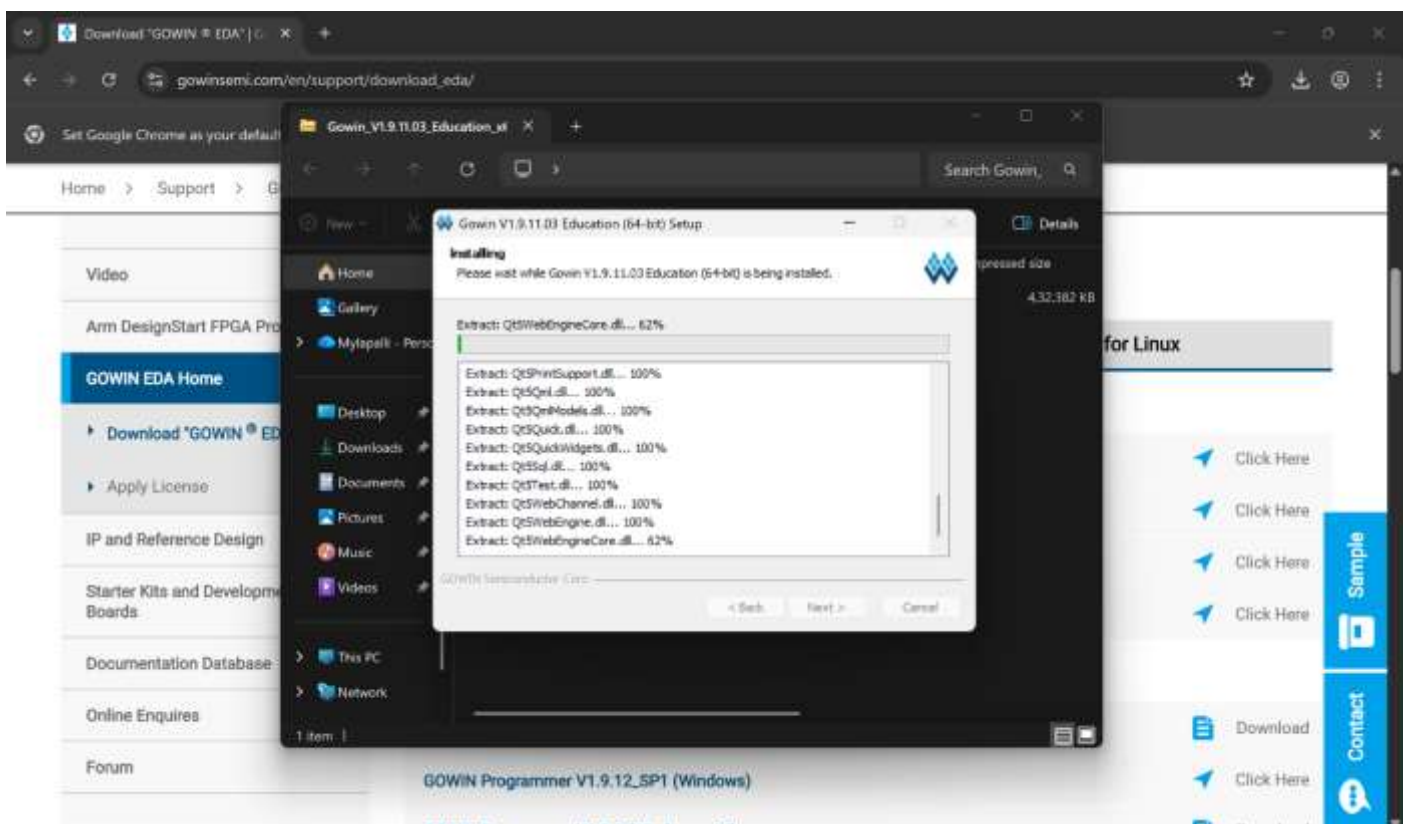
After clicking the Application, we will be having a welcome dialogue box with 2 options those are next and Cancel. Choose next option in that



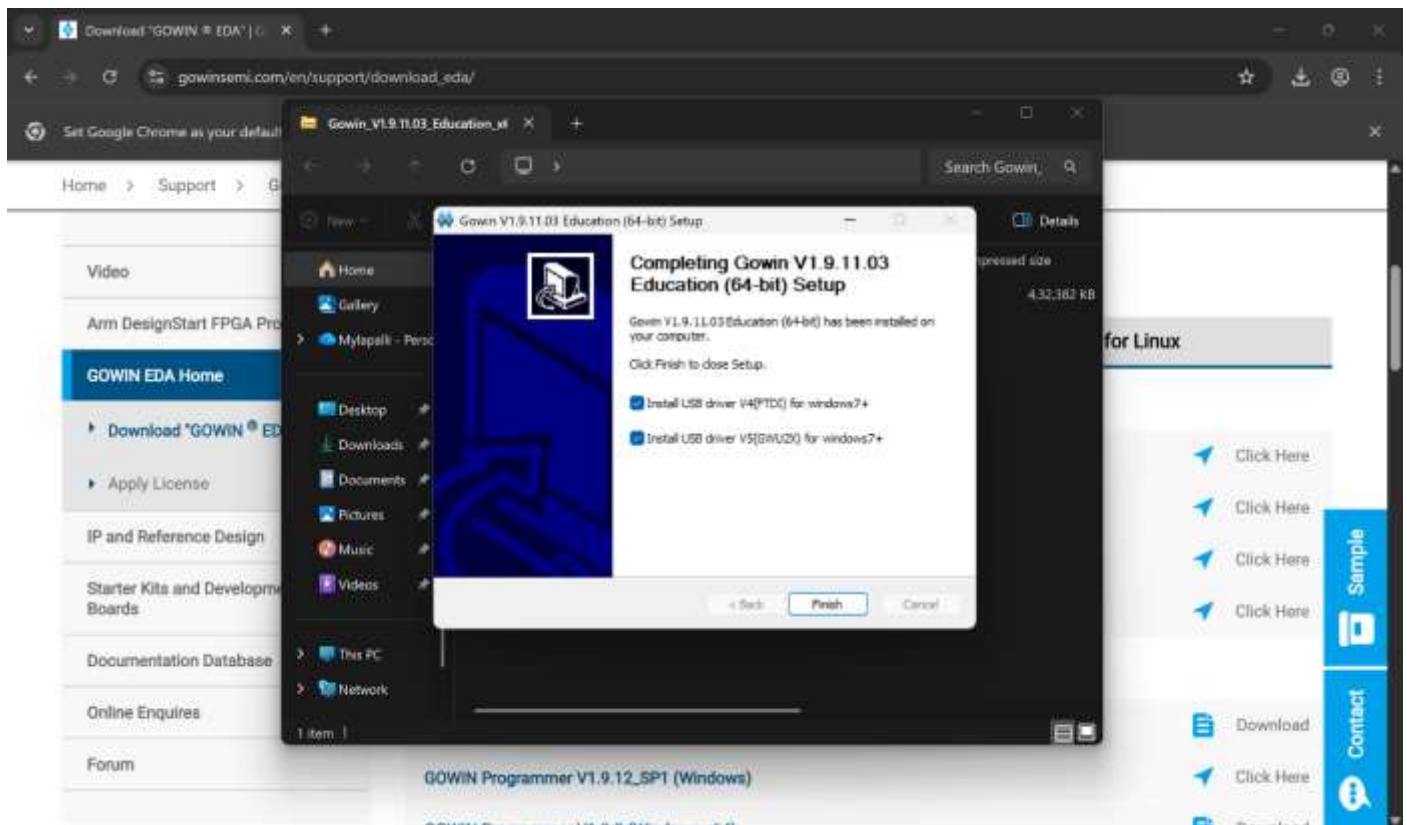
After clicking Next, Click on the I AGREE option in the appeared dialogue box



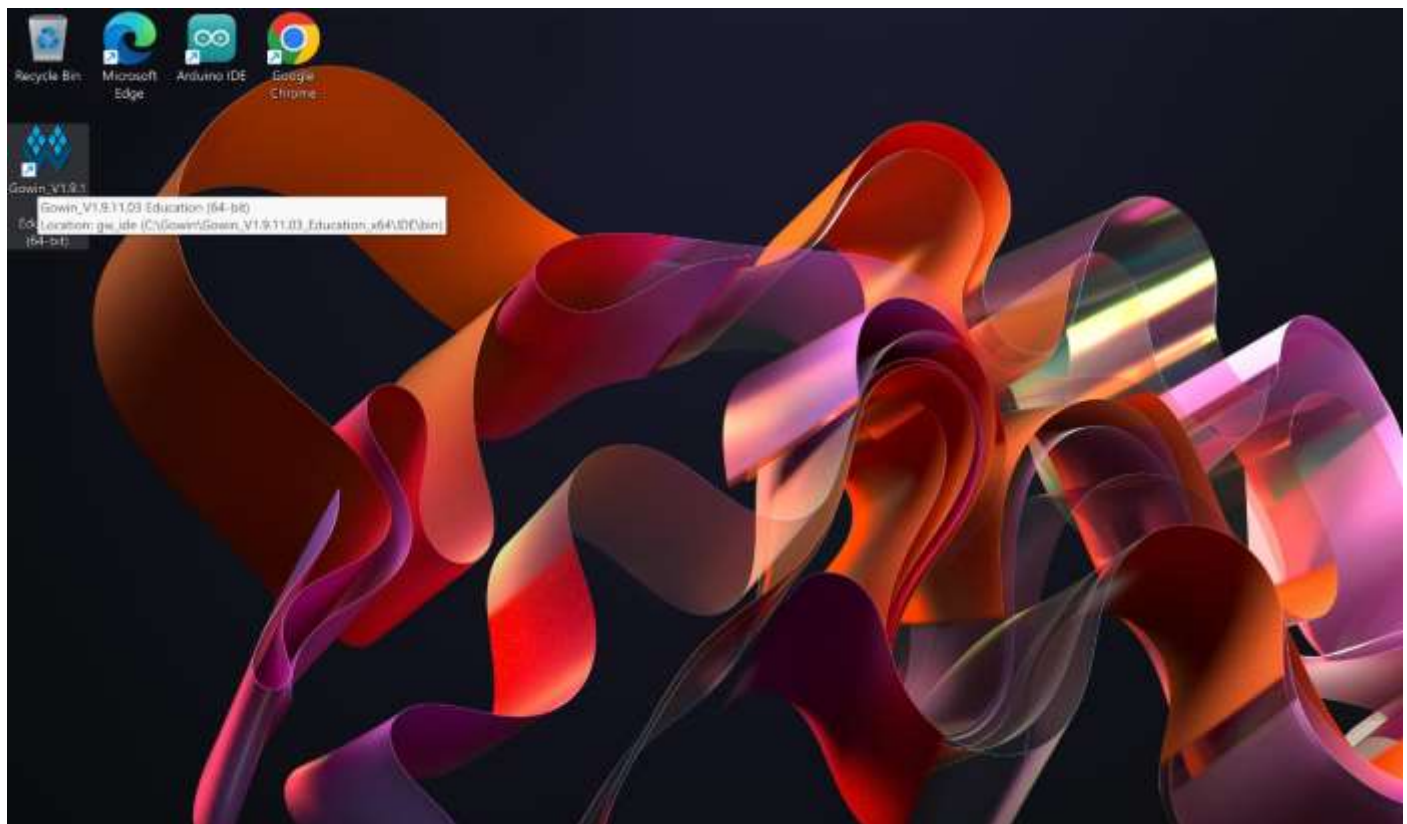
After that another box will be appeared, click on the NEXT option in that



After that all, we will be seeing that GOWIN IDE will be installed in our PC.



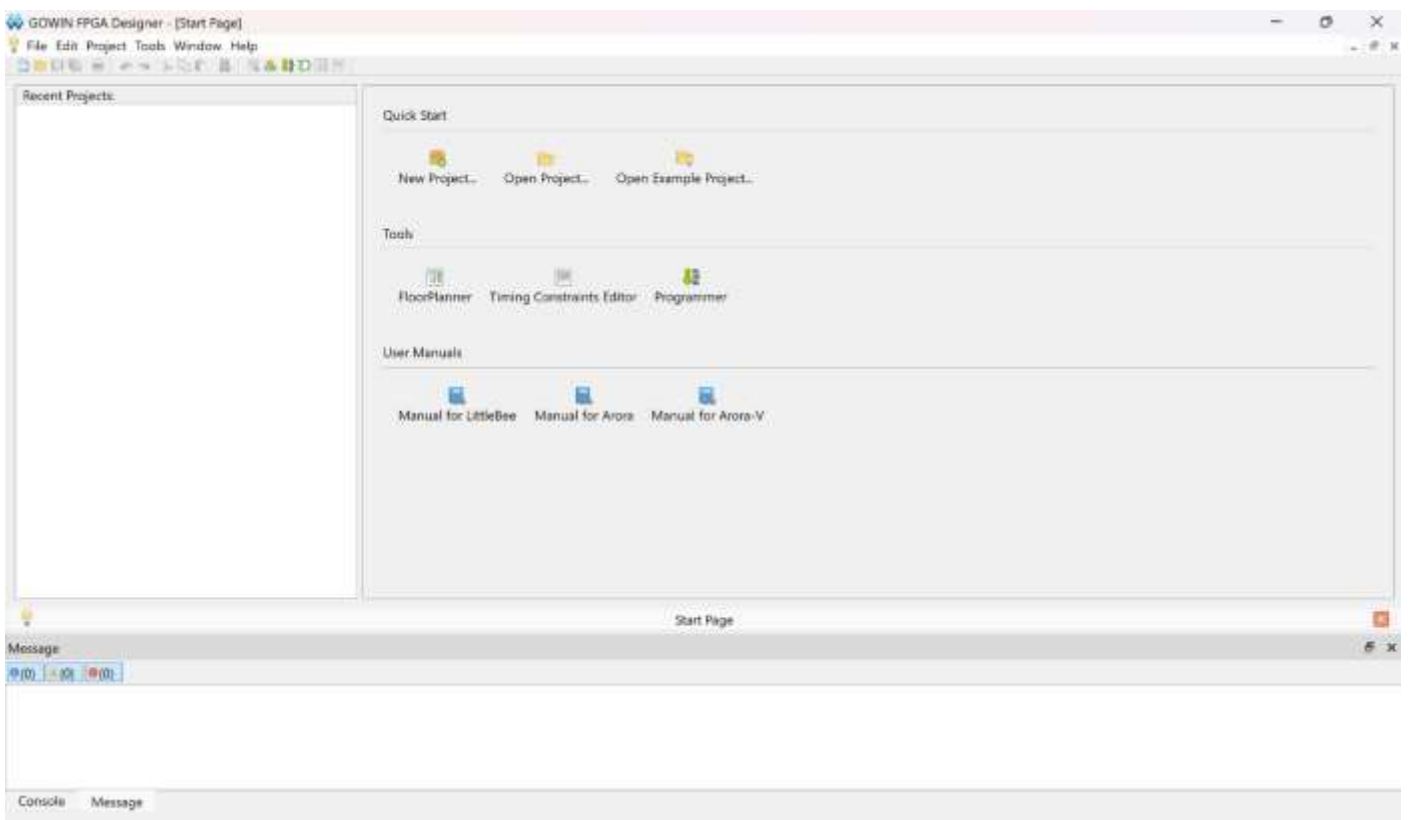
we will see a box after installing the IDE like the above picture. Click on the finish button



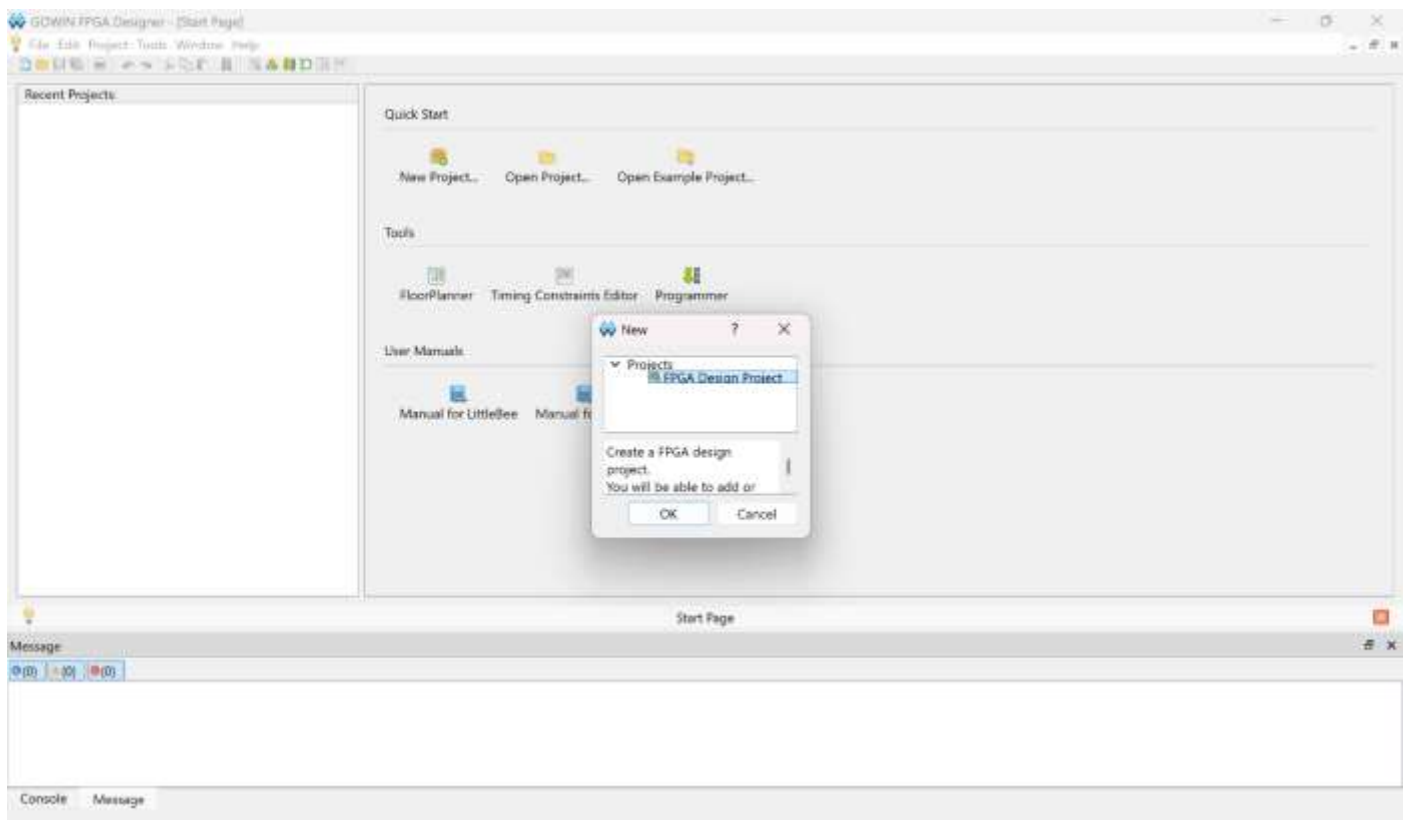
After all the above process we will have the GOWIN IDE in our Desktop.



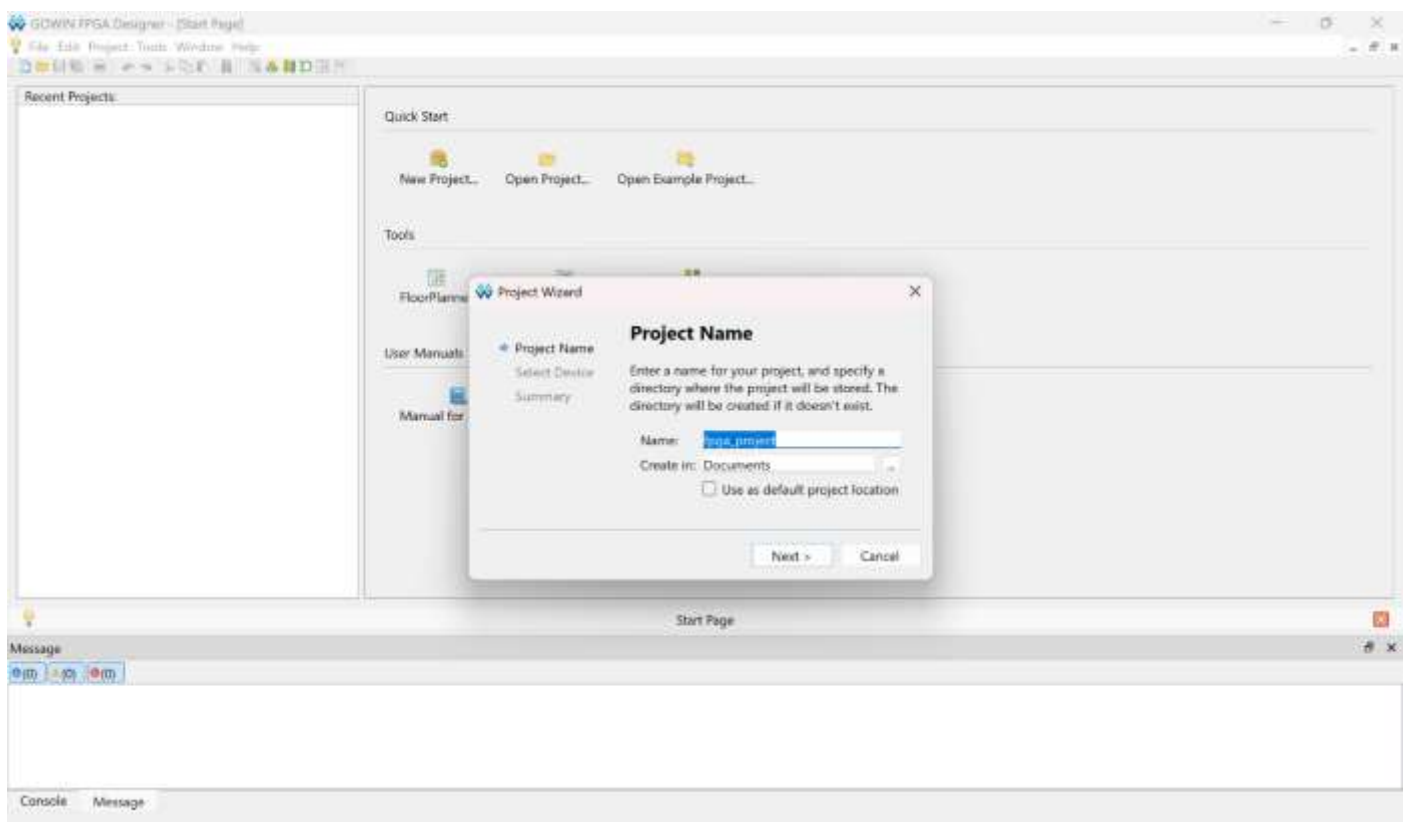
Open the GOWIN IDE .



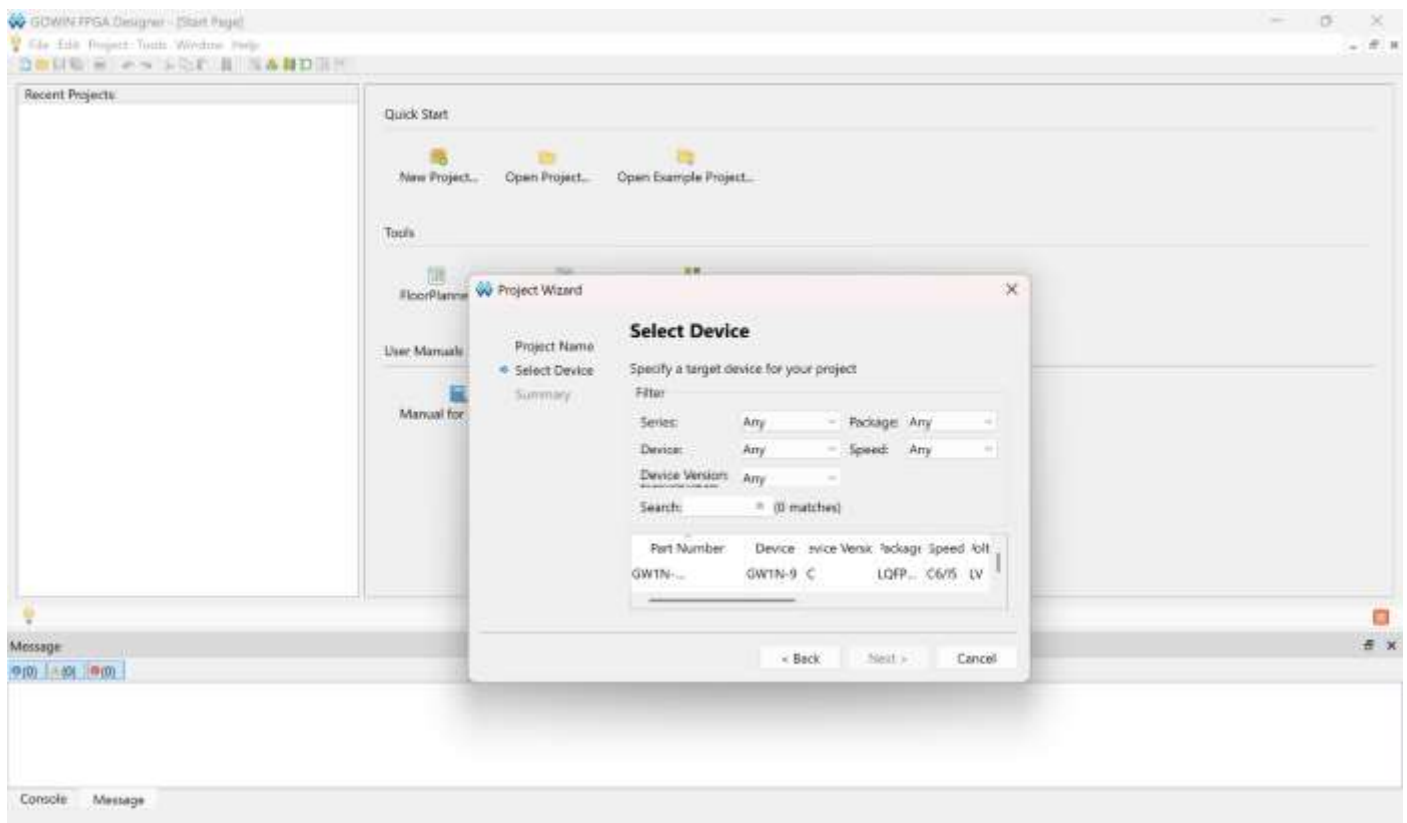
After opening ide we will having the interface like this. Click on the new project in it.



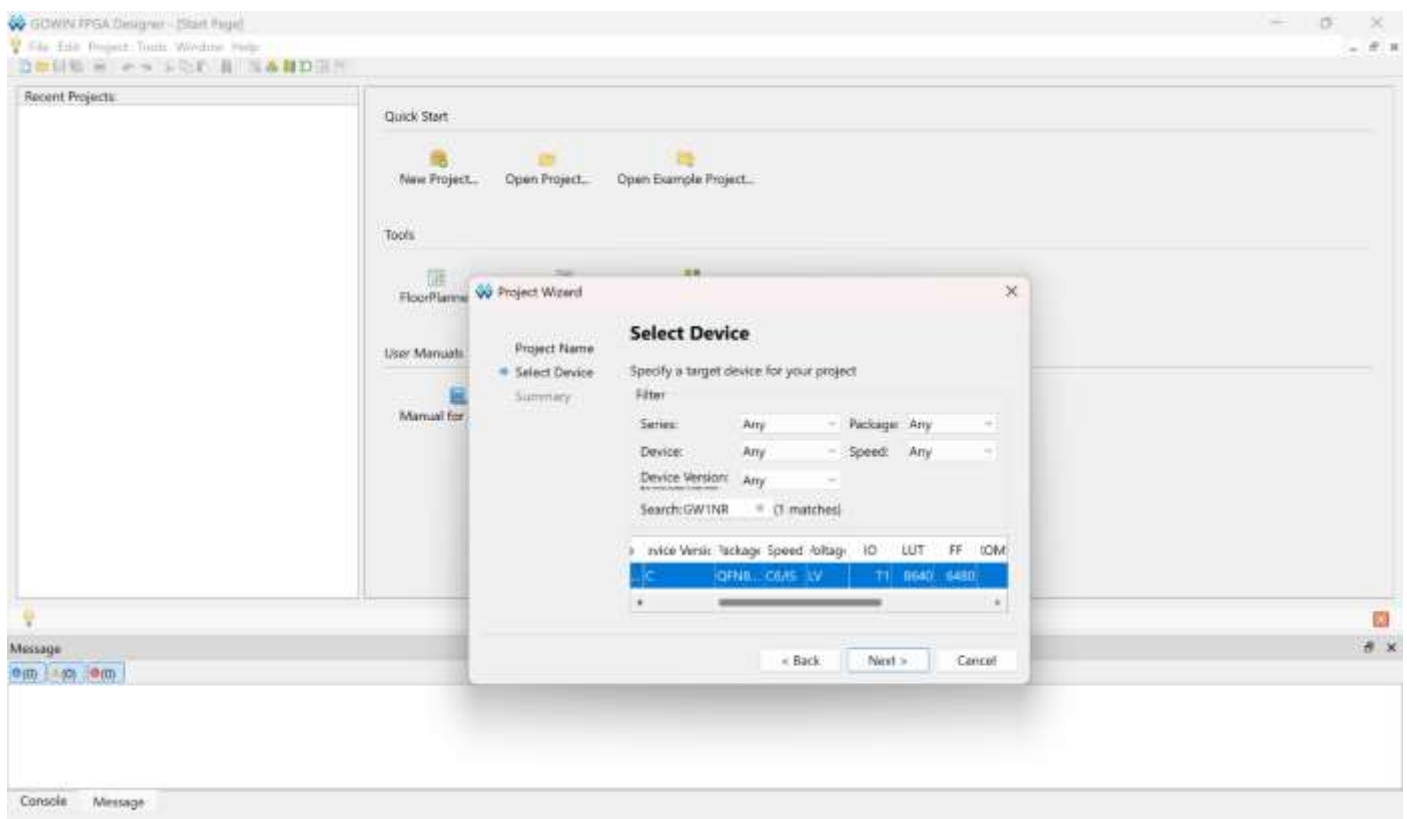
After clicking new project, select FPGA Design Project and click OK.



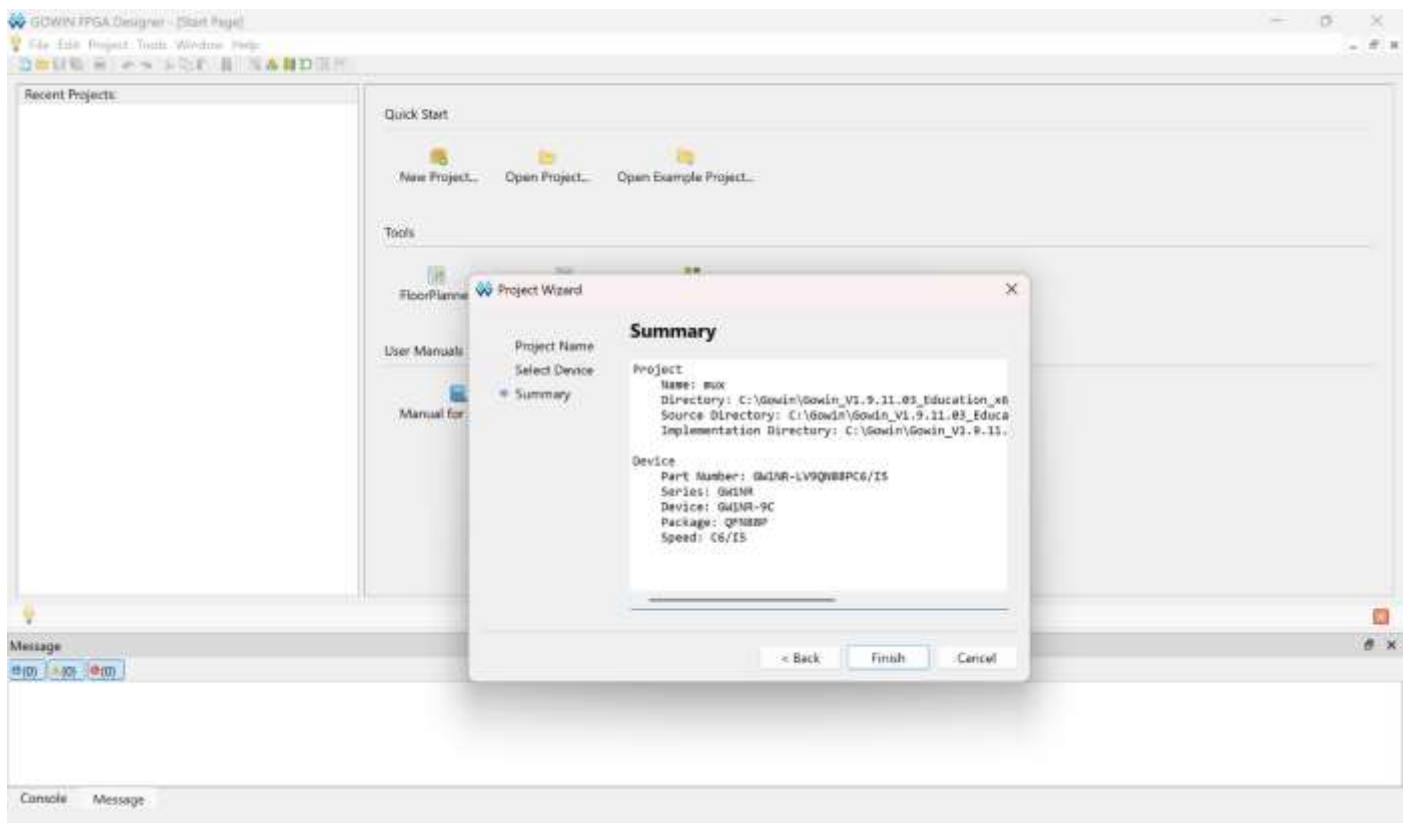
Name the project as we want. here we kept “mux”



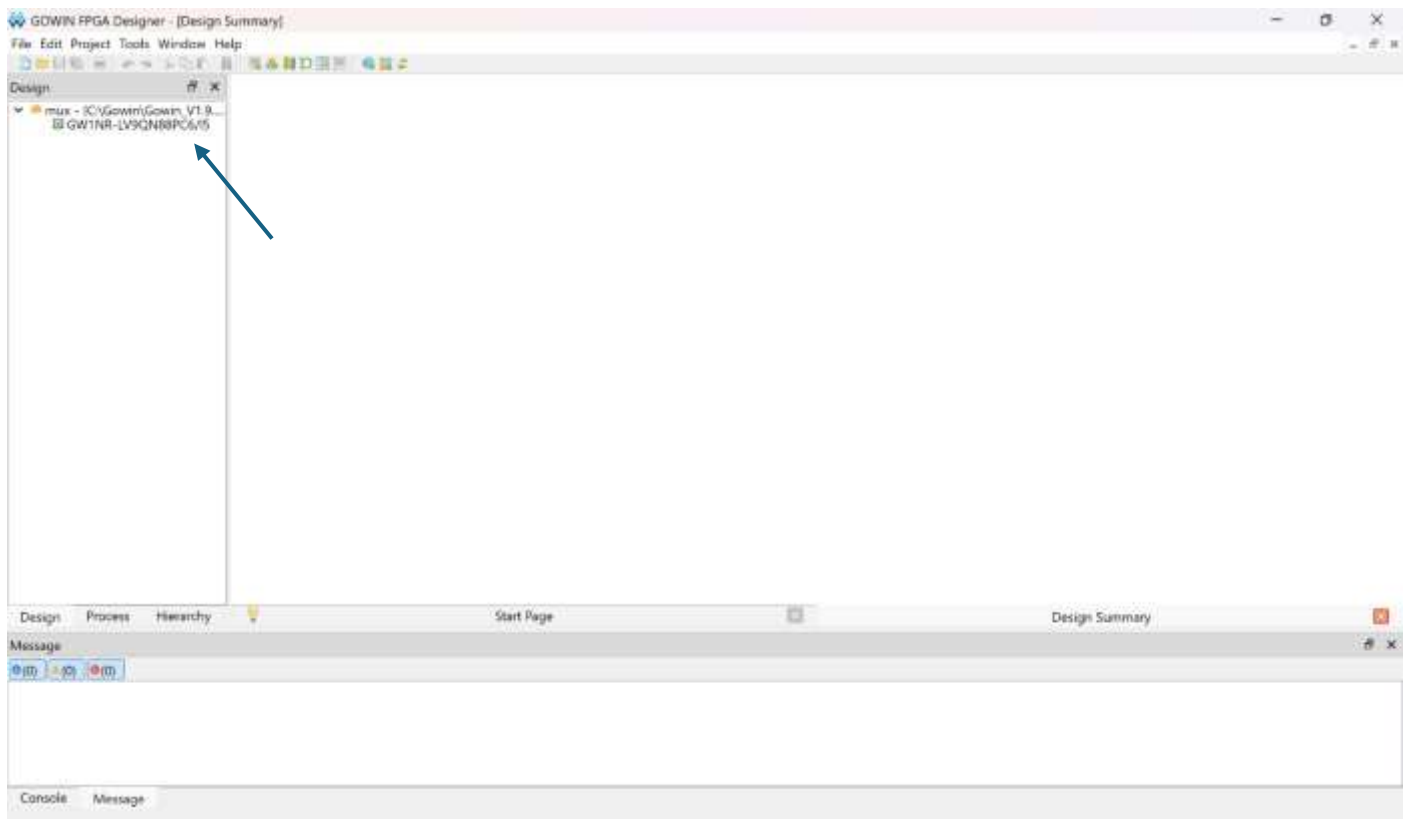
After naming the project we will have a box, that asks about the Device we used and its having all the details like Part number, Device, Version, Speed etc. select the GW1NR-9C with speed C6/15 of package LQFP88P



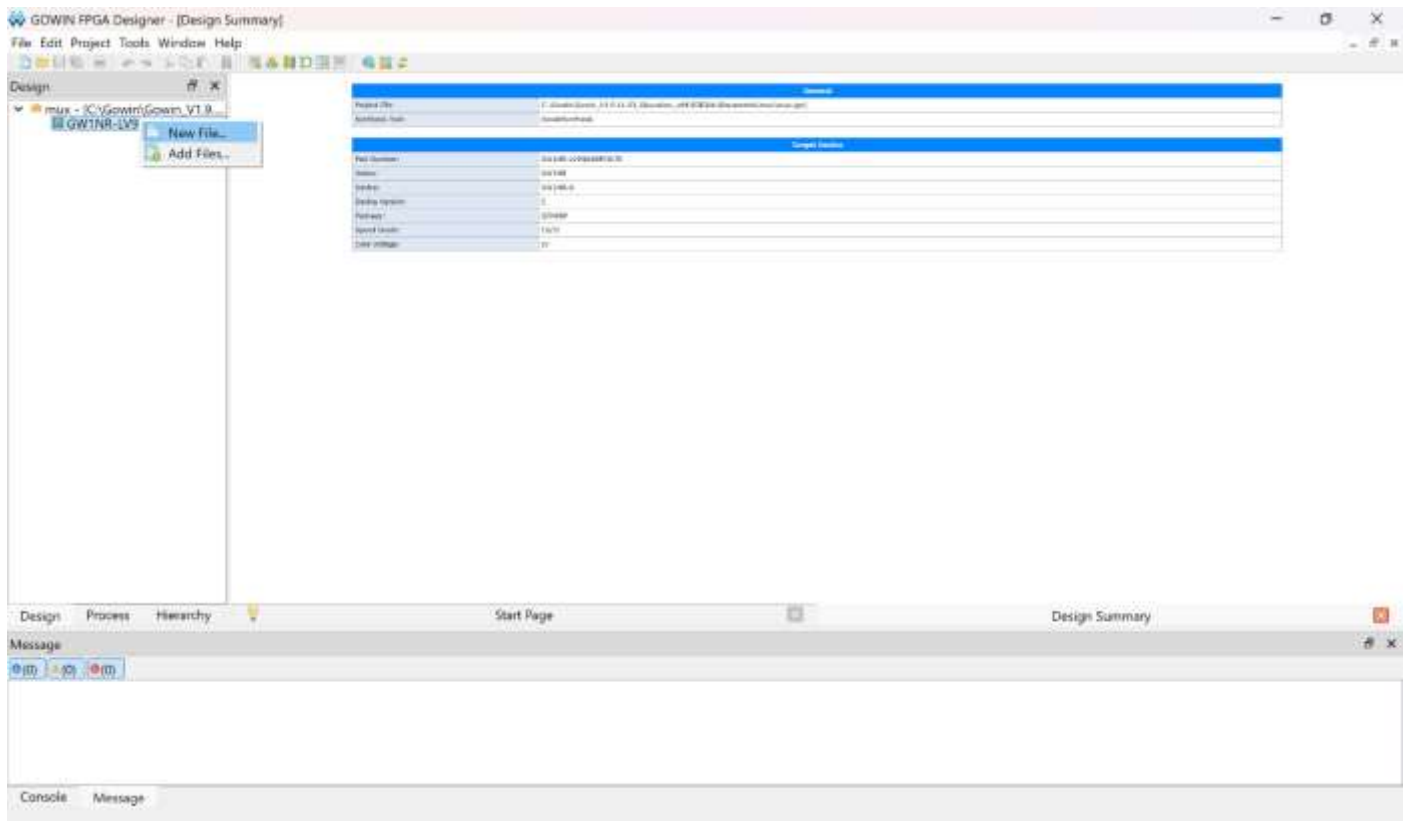
Select the required FPGA device and Select the device that we are supposed to use. select the GW1NR-9C with speed C6/15 of package LQFP88P



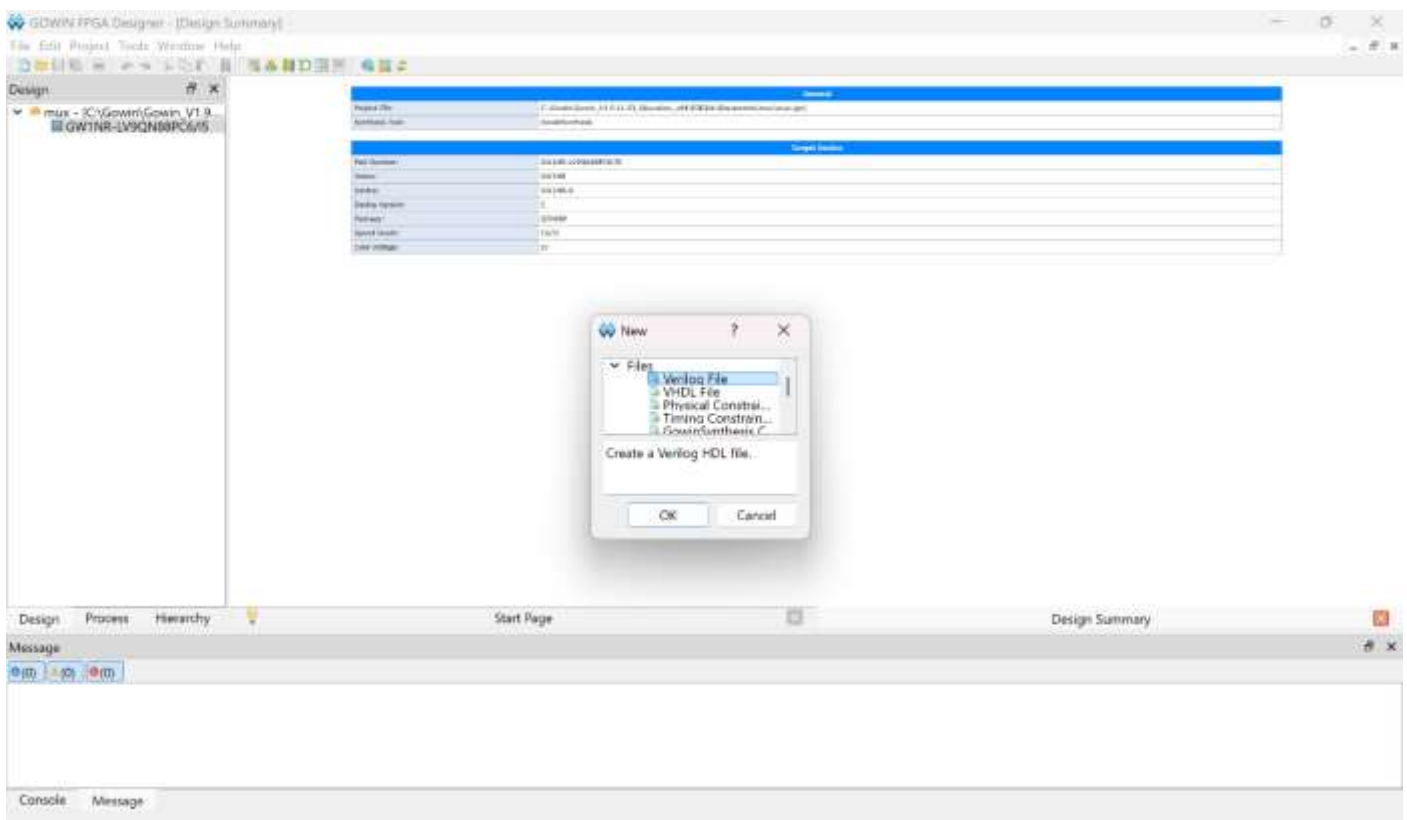
After selecting the device we will get all the details of the device we selected. Click on the finish button.



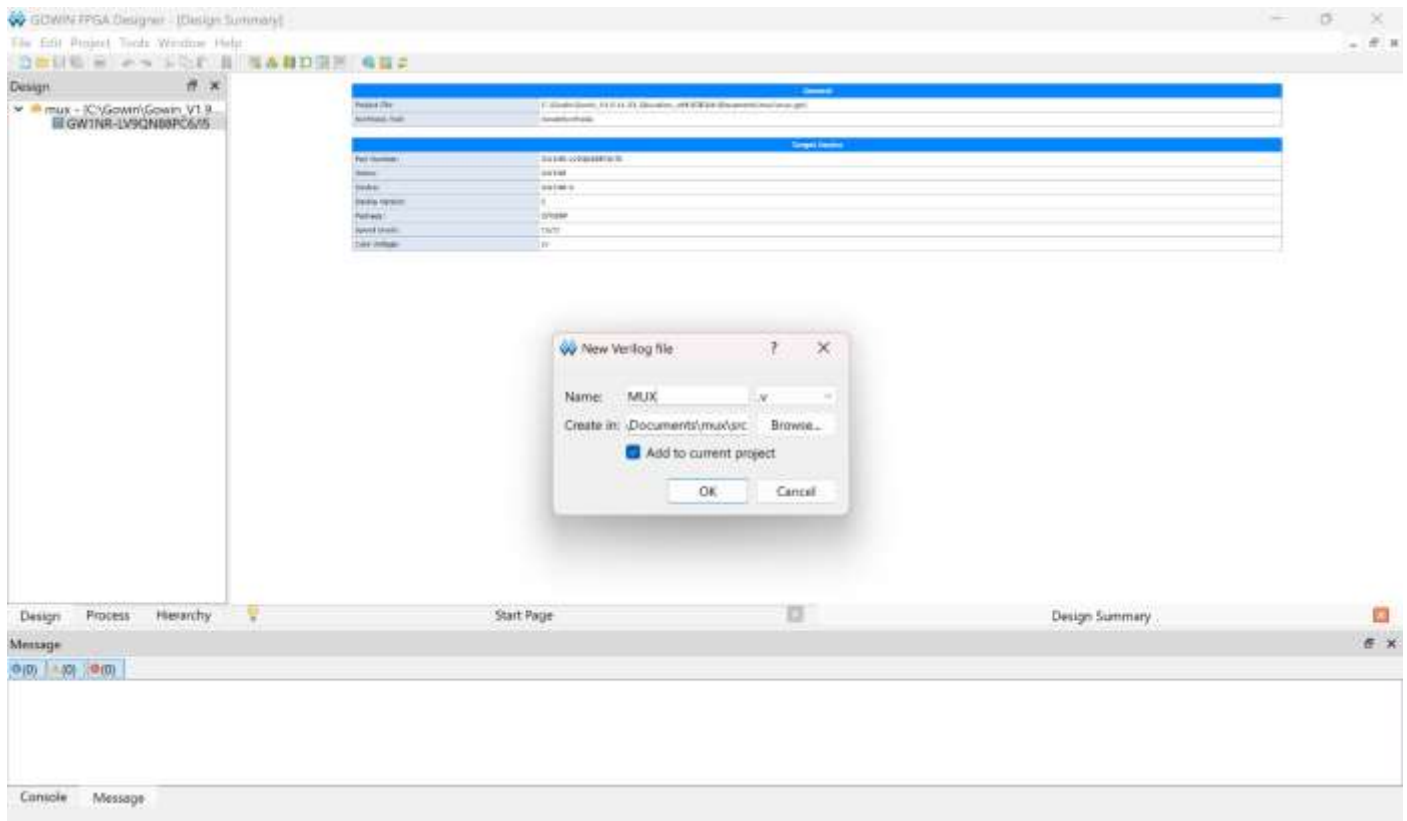
After clicking the finish button we will be directed to workspace where we have to write Verilog code. Now click on the file below the mux folder as shown in the picture.



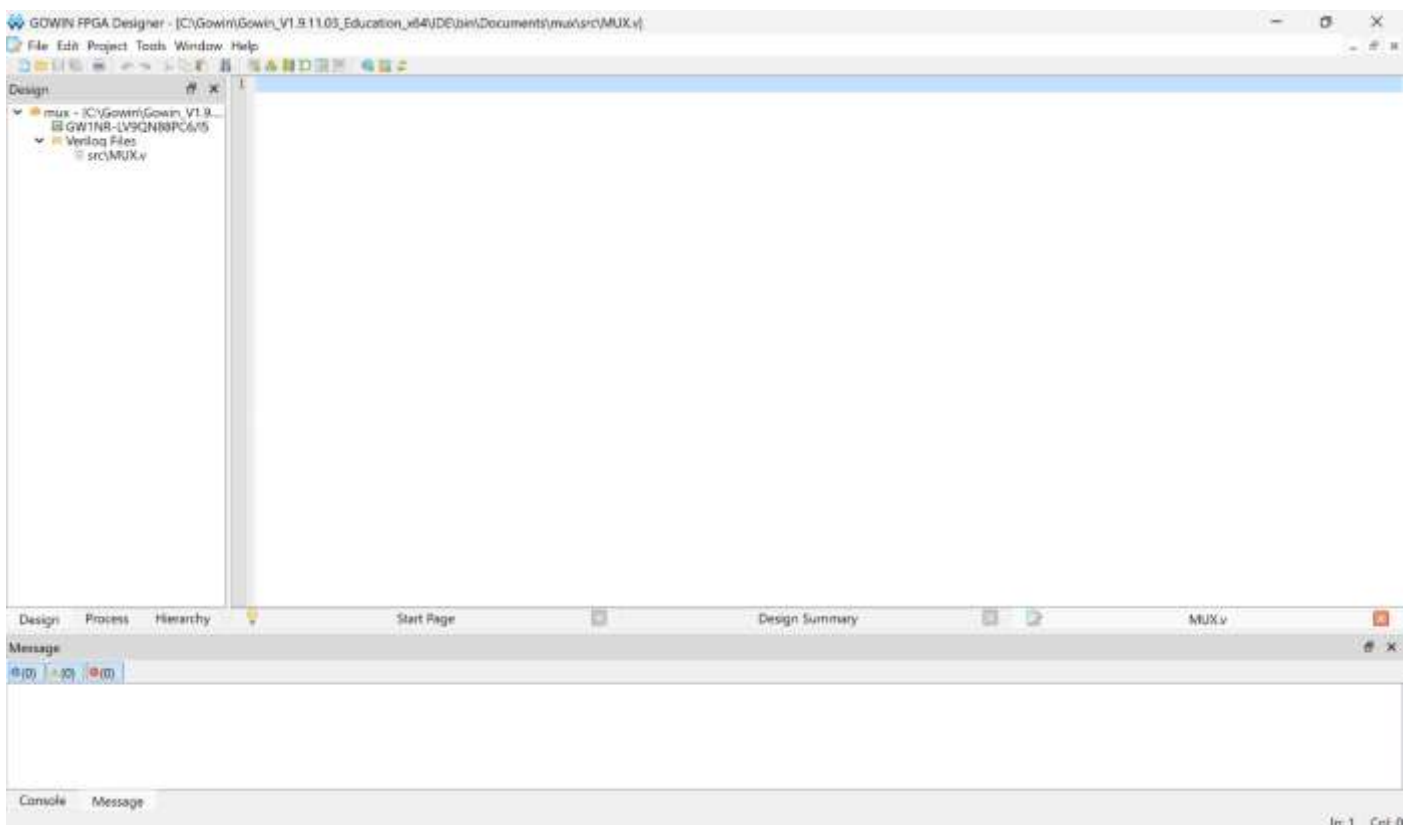
We will have 2 options: those are new file and add file. Click on the new file.



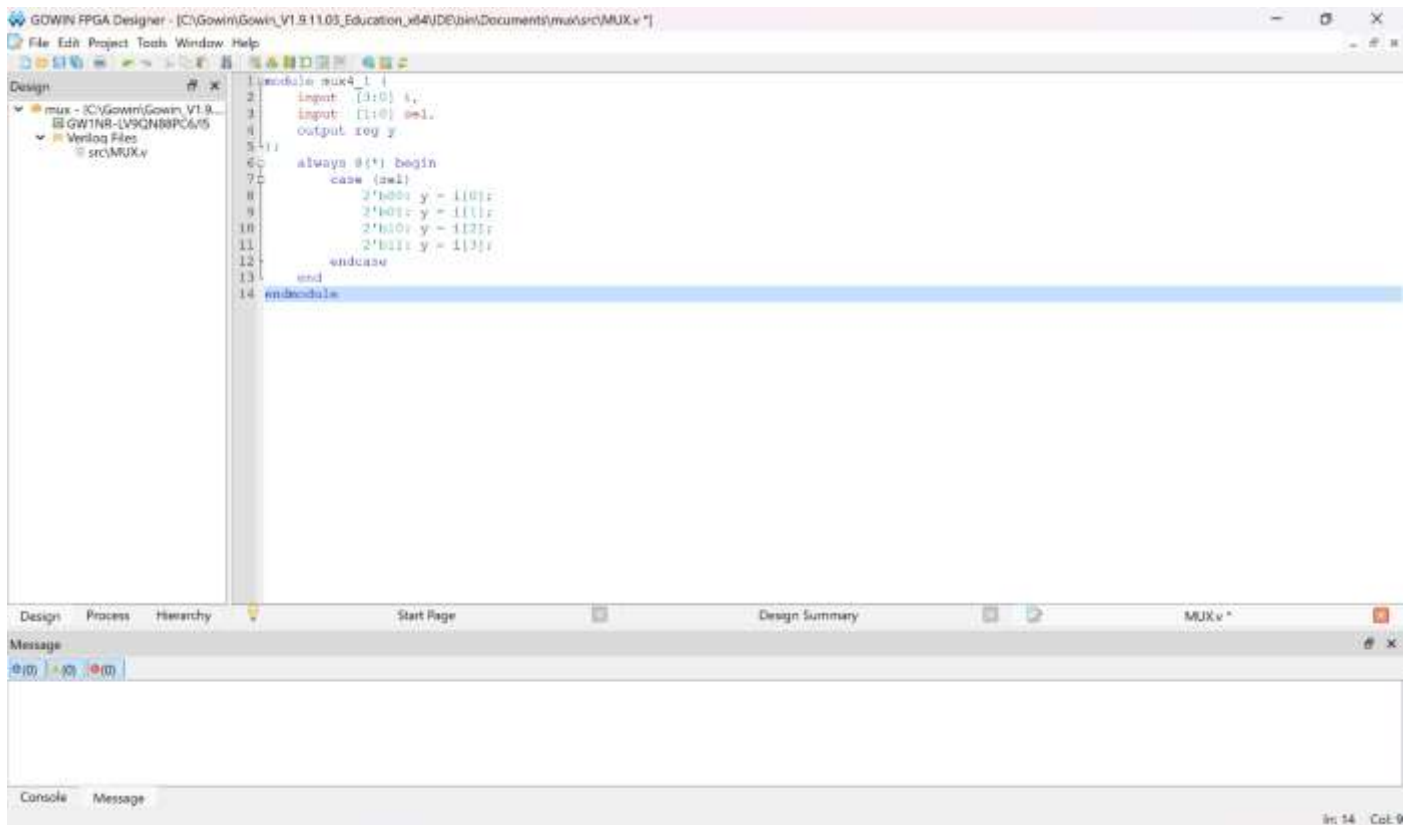
After clicking the new file we will have many different files like Verilog file, VHDL file etc. click on the Verilog file.



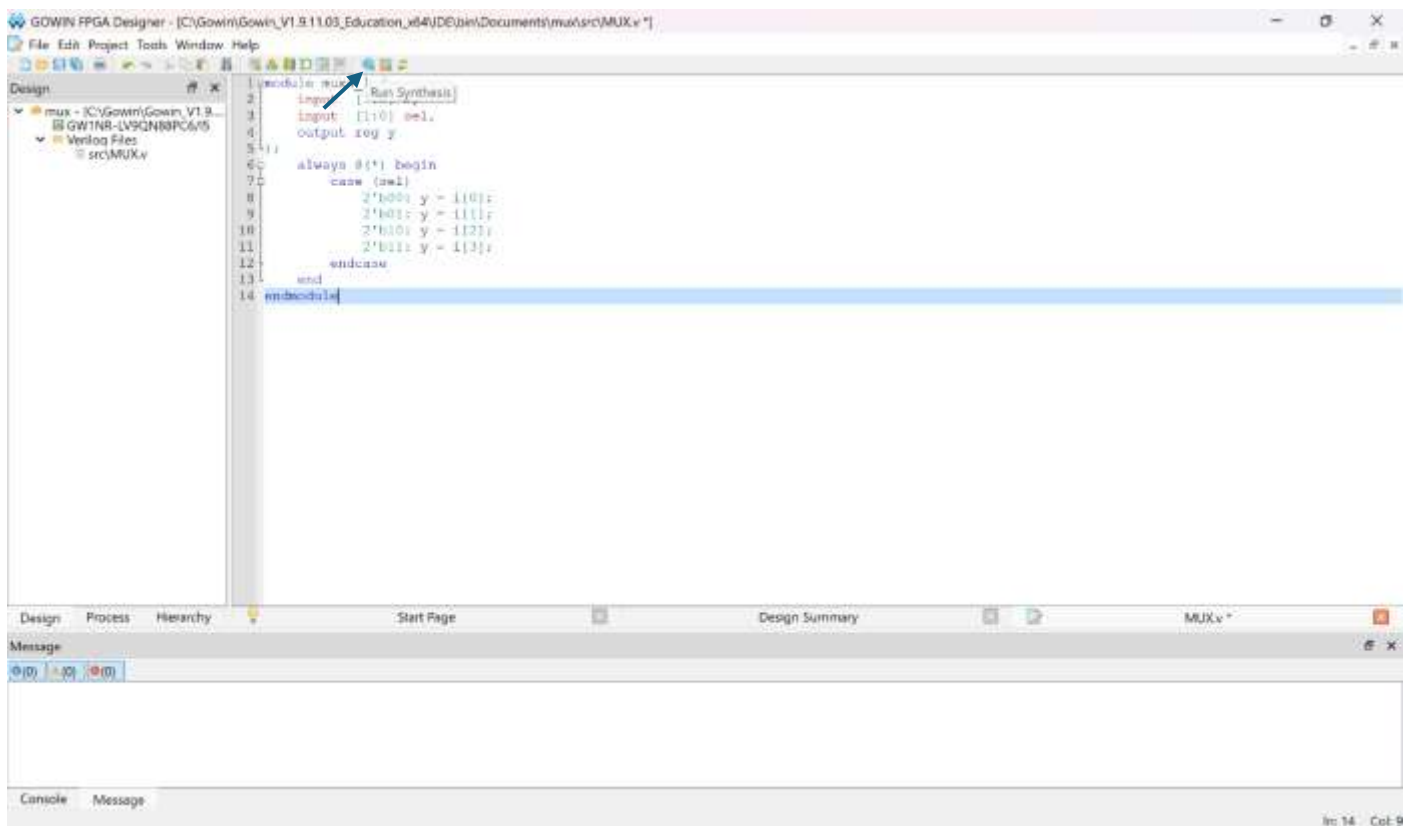
After selecting Verilog file, name the Verilog file and select the OK option in it.



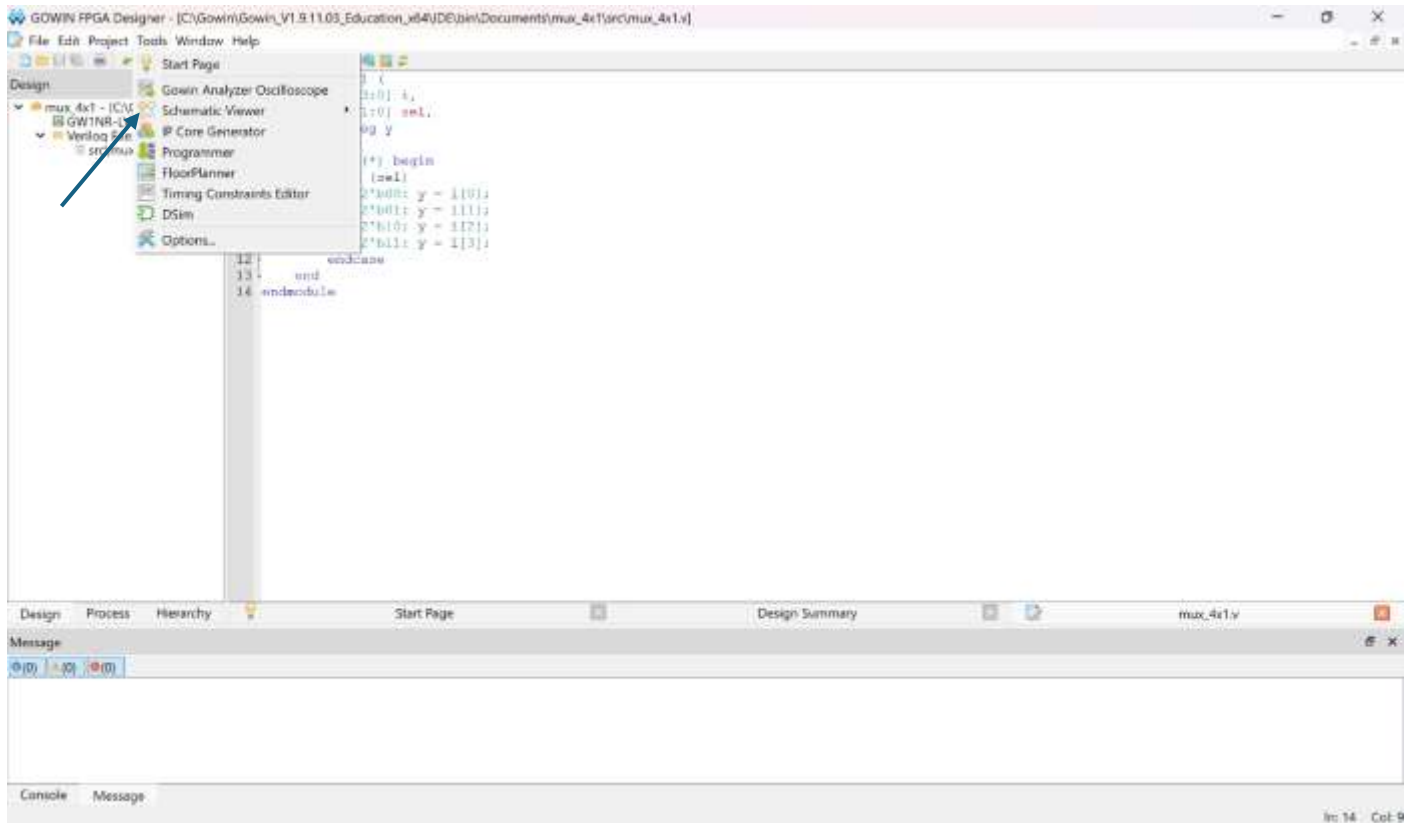
After naming and clicking OK we will be having a free space, where we have to write Verilog code.



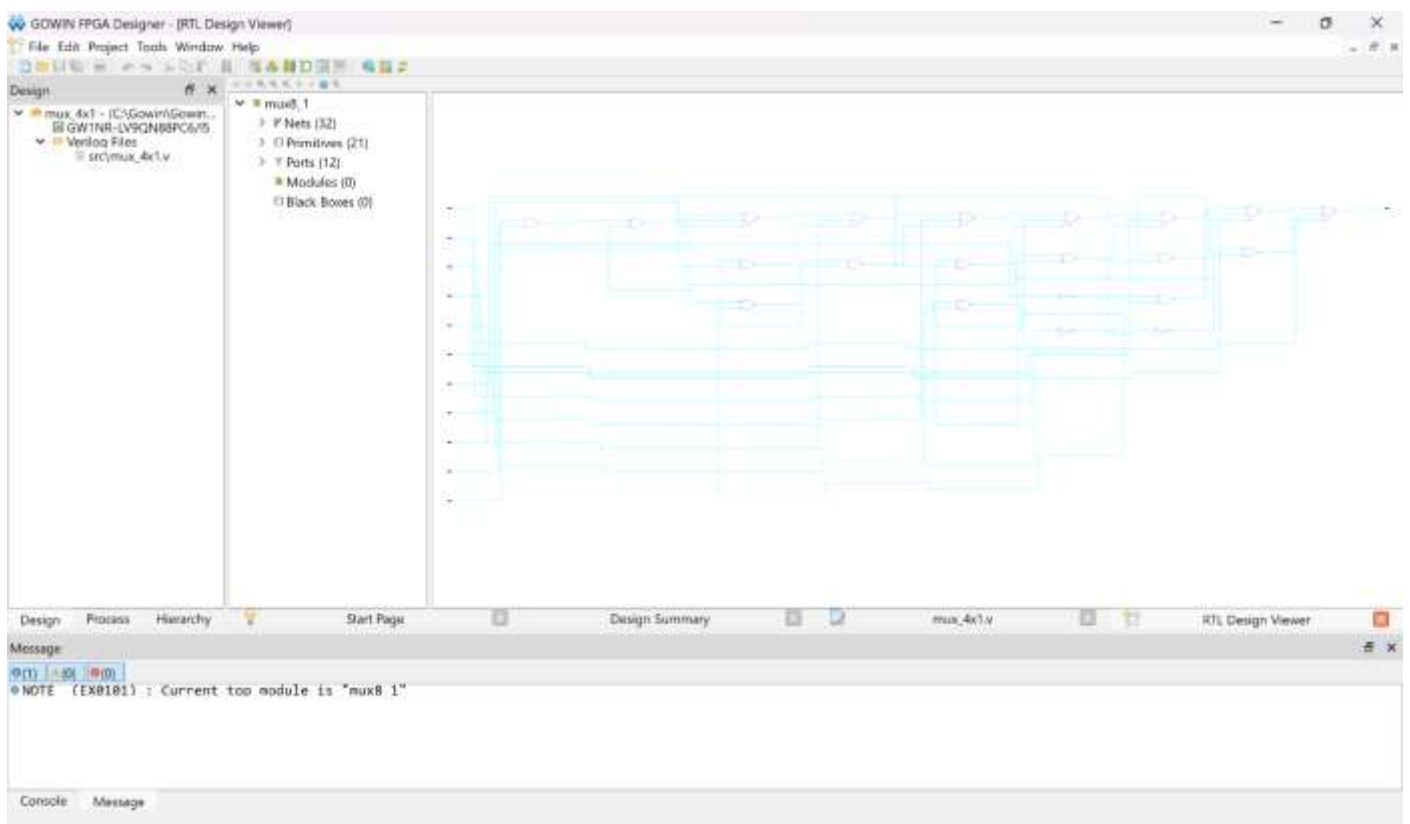
Write the code that we want to implement on the FPGA board.



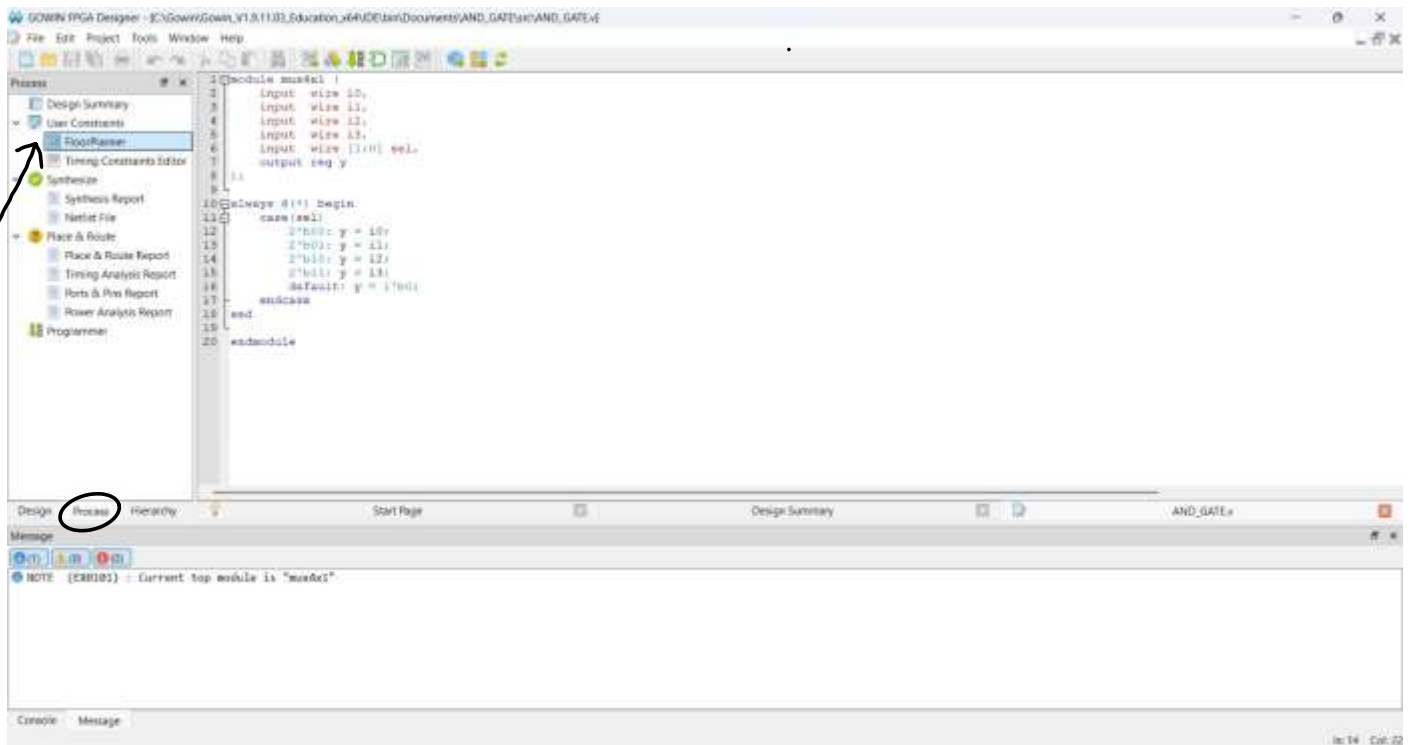
After writing the code click on the RUN SYNTHESIS option as shown in the picture.



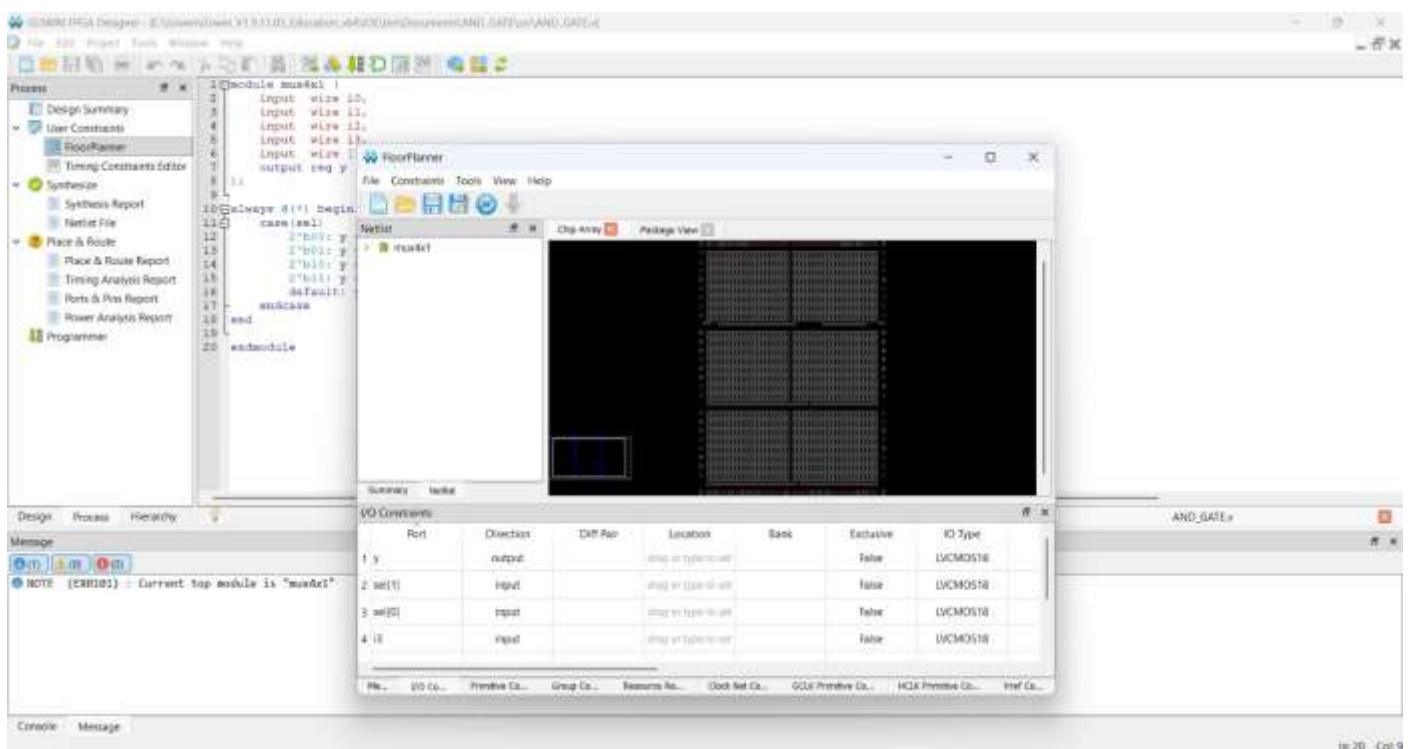
After clicking run synthesis, click on the tools option and select the schematic viewer .



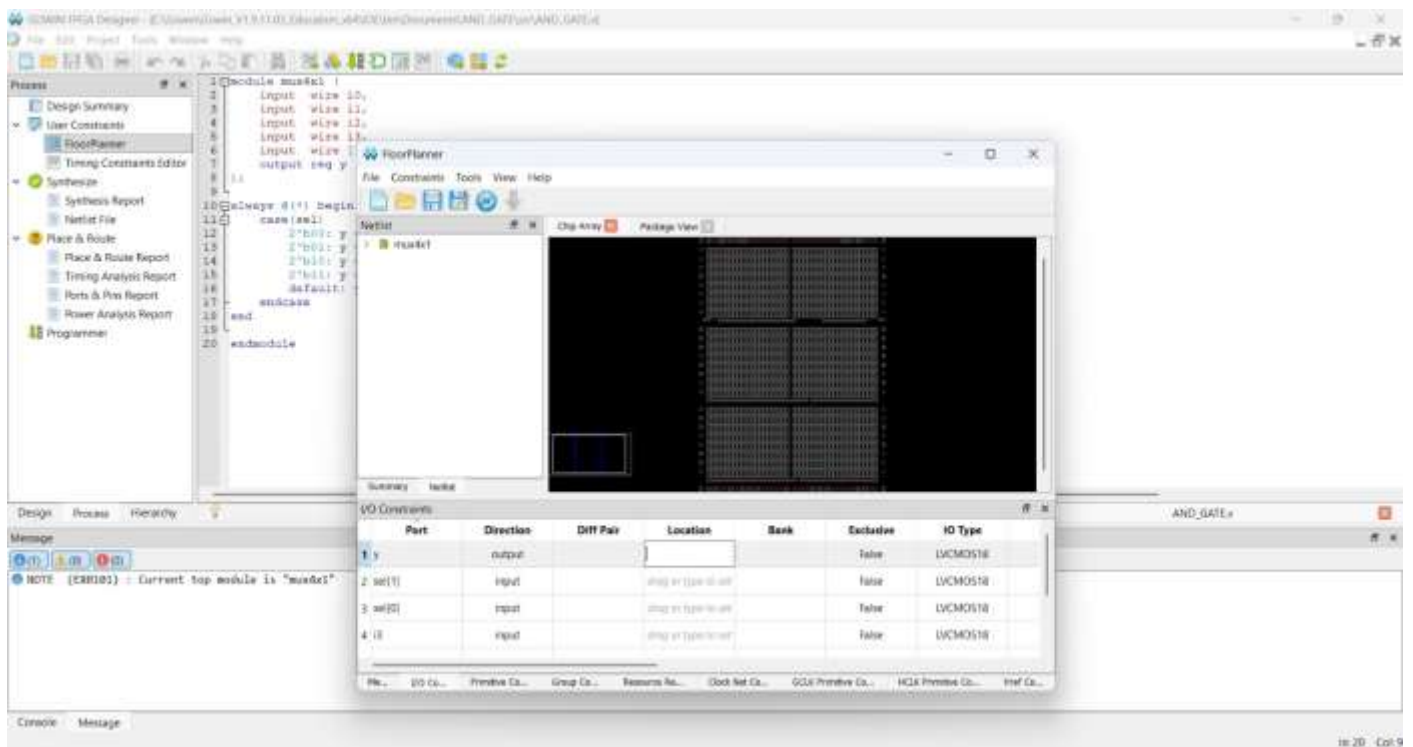
After clicking the schematic viewer we can see the circuit diagram of the Multiplexer .



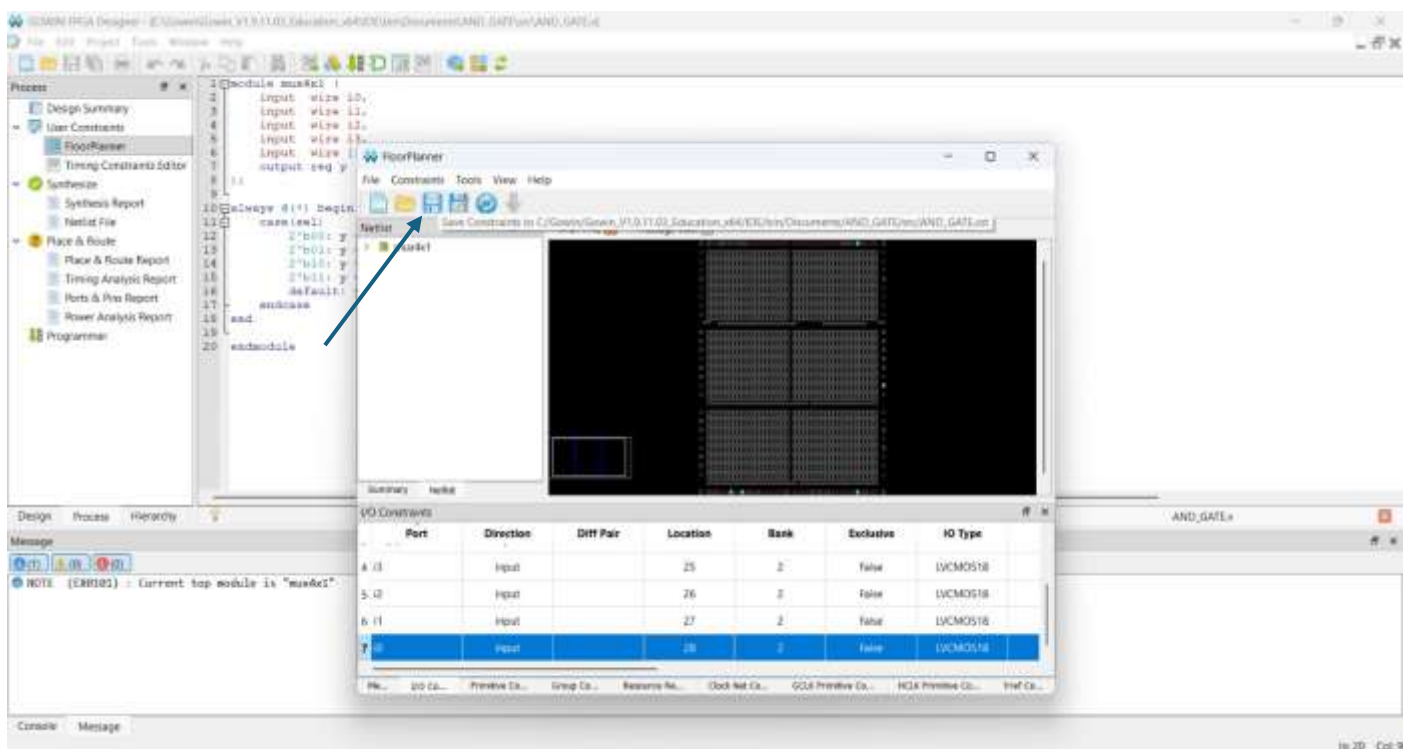
Click on the process button and select the floor planar option in it.



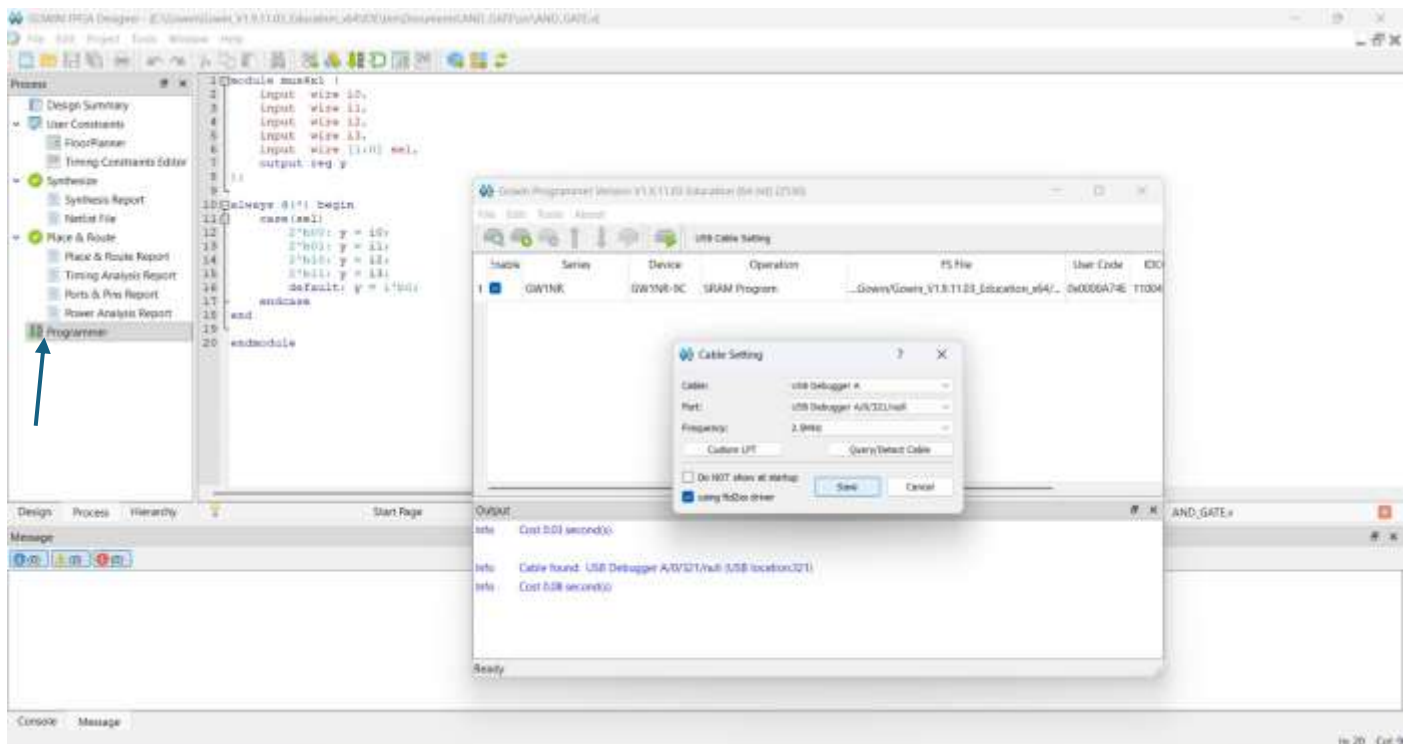
After clicking on the floor planar, we can see the above dialogue box that shows the internal architecture of the FPGA.



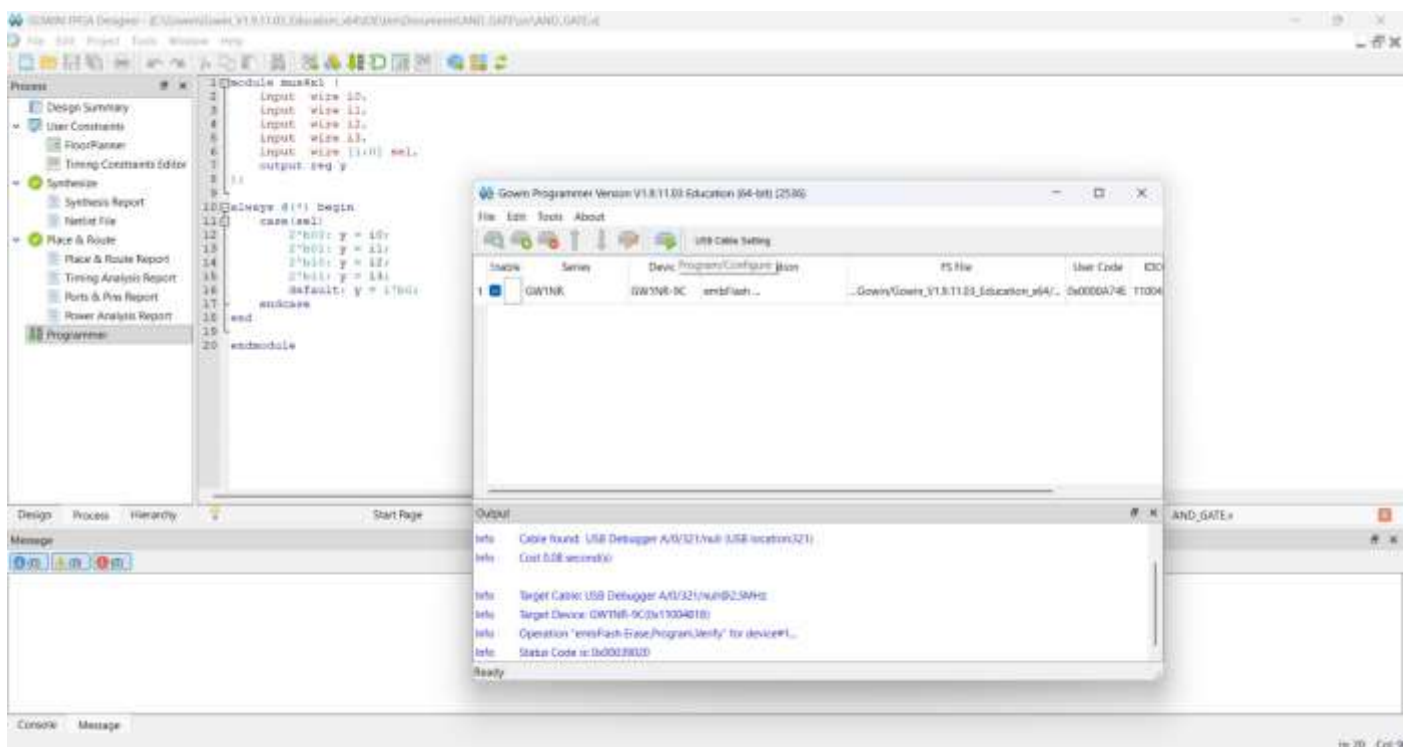
Here we need to assign the inputs of the circuits to the SWITCHES and output of the circuit to the LEDs by entering the input switch numbers on trainer kit and output leds on the trainer kit



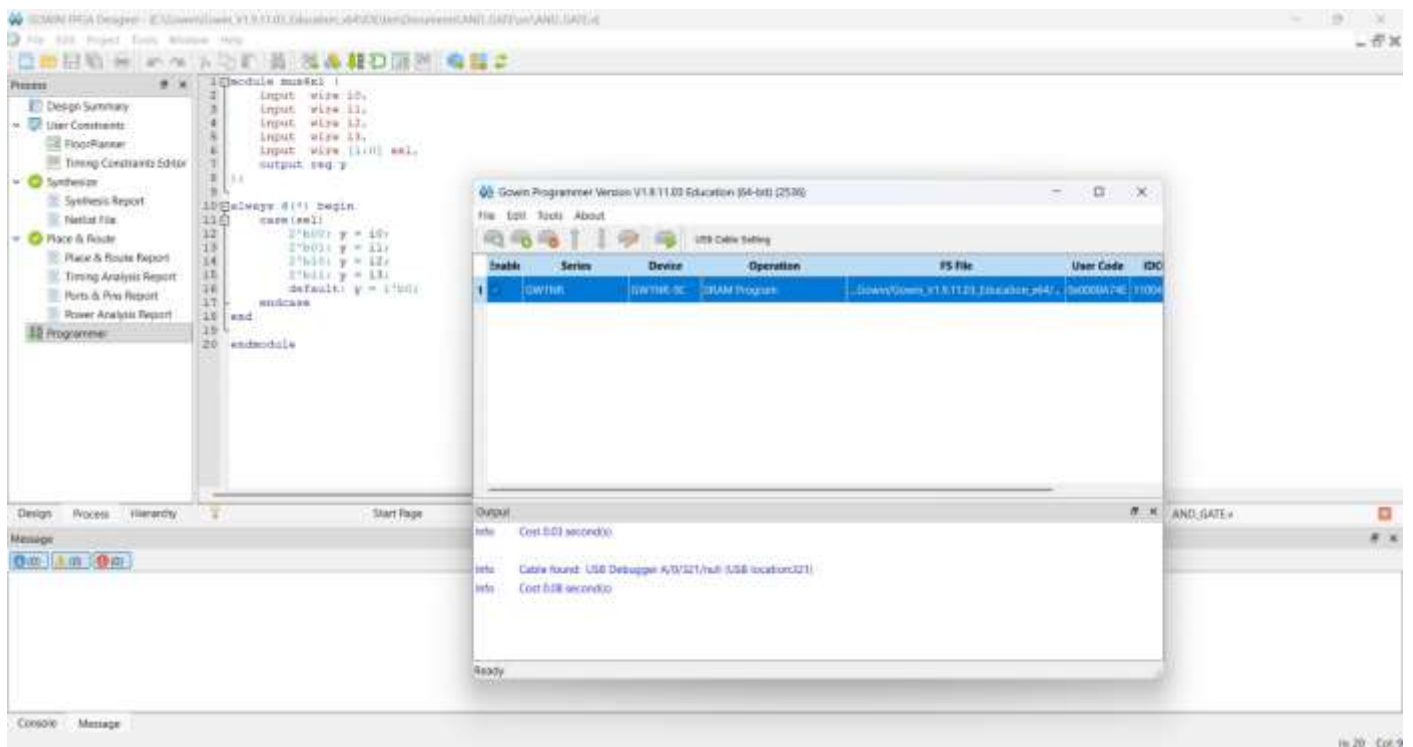
After assigning the inputs and outputs in the floor planner, click on the save option



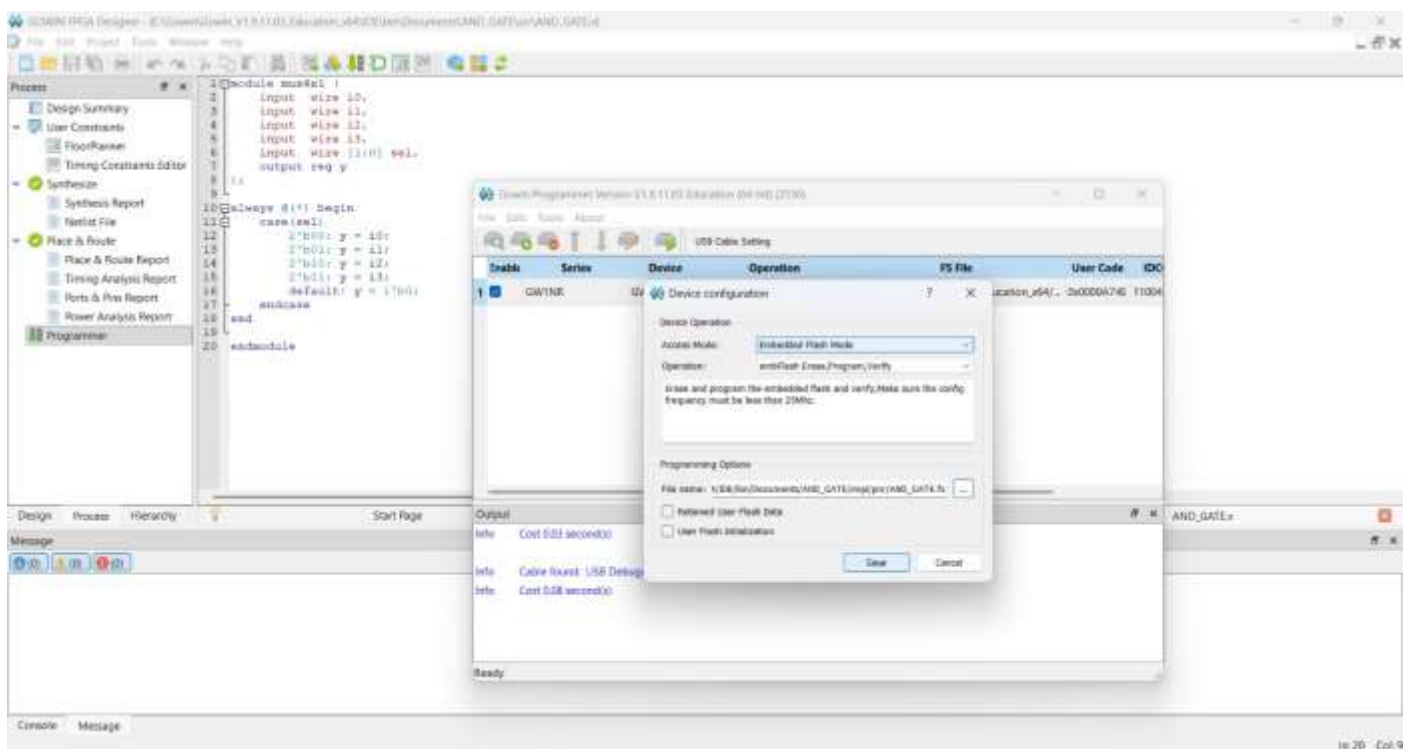
now click on the programmer , the above box will appear with many options like cable and port settings. We need to connect our fpga with pc before entering into the programmer, when the fpga is detected, the programmer will automatically detect the cable and port with the required frequency. After detection click on the save button.



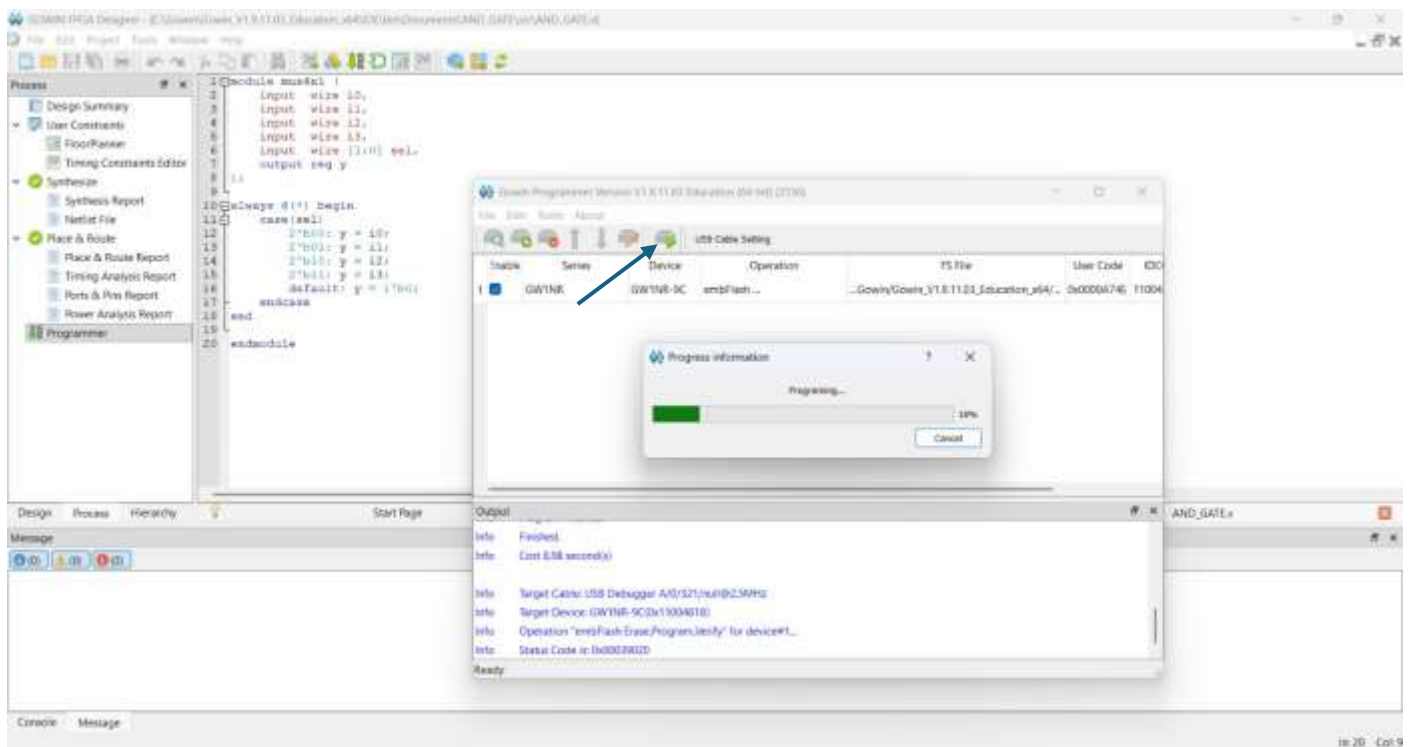
Now select the device that shows on the programmer dialogue box.



Select the device by clicking on it.



In device configuration, select the access mode with Embedded flash Mode and operation with embflash erase,program,verify. And click on the save button.



After fixing and selecting required modes..then click on the upload button and start uploading.

We can implement any of Digital Circuit on this FPGA using GOWIN IDE only and for GW1NR Device Only.